

1 SuSiE (Sum of Single Effects)

1.1 Simulation Design

Wang et al. simulate data designed to mimic realistic genetic fine-mapping of expression quantitative trait loci (eQTLs). The design uses *real* genotype matrices combined with *synthetic* phenotype vectors.

1.1.1 Genotype Data (Design Matrix $\mathbf{X}$)

Individual-level genotype data are taken from $n = 574$ samples collected as part of the Genotype–Tissue Expression (GTEx) project. To simulate *cis*-eQTL fine mapping, 150 genes are randomly selected from over 20 000 genes on chromosomes 1–22. For each gene, the design matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ consists of genotypes at SNPs near the transcribed region of the selected gene, with the number of SNPs ranging from $p = 1\,000$ to $p = 12\,000$ depending on the gene and simulation scenario. Further details on SNP selection criteria are given in the supplementary appendix D of the original paper.

These real genotype matrices exhibit complex patterns of linkage disequilibrium (LD). For a randomly chosen variable, the median number of other variables with which its absolute correlation exceeds 0.9 is 8, and the median number exceeding 0.99 is 1 (with corresponding means of 26 and 8, respectively, due to some variables being strongly correlated with hundreds of others).

1.1.2 Generative Model for the Phenotype

The phenotype vector $\mathbf{y} \in \mathbb{R}^n$ is generated under the standard multiple-regression model:

$$\mathbf{y} = \mathbf{X}\mathbf{b} + \mathbf{e}, \quad \mathbf{e} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n), \quad (1)$$

where $\mathbf{b} \in \mathbb{R}^p$ is a sparse vector of regression coefficients and $\sigma^2 > 0$ is the residual variance. The simulation of $\mathbf{b}$ and $\mathbf{y}$ is governed by two parameters:

- S : the number of causal (effect) variables;
- ϕ : the proportion of variance in $\mathbf{y}$ explained (PVE) by $\mathbf{X}$.

Given S and ϕ , the data-generating process proceeds as follows:

1. **Select causal variants.** Sample the indices of the S effect variables, denoted $\mathcal{S}$, uniformly at random from $\{1, \dots, p\}$.

2. **Draw effect sizes.** For each $j \in \mathcal{S}$, independently draw

$$b_j \sim \mathcal{N}(0, 0.6^2), \quad (2)$$

and for all $j \notin \mathcal{S}$, set $b_j = 0$.

3. **Set residual variance.** Determine σ^2 to achieve the desired PVE ϕ by solving:

$$\phi = \frac{\text{var}(\mathbf{X}\mathbf{b})}{\sigma^2 + \text{var}(\mathbf{X}\mathbf{b})}, \quad (3)$$

where $\text{var}(\cdot)$ denotes the sample variance. That is,

$$\sigma^2 = \text{var}(\mathbf{X}\mathbf{b}) \left(\frac{1 - \phi}{\phi} \right). \quad (4)$$

4. **Simulate phenotype.** For each $i = 1, \dots, n$, draw

$$y_i \sim \mathcal{N}\left(\sum_{j=1}^p x_{ij} b_j, \sigma^2\right). \quad (5)$$

1.1.3 Simulation Scenarios

Two simulation scenarios are considered:

Scenario 1 ($p = 1\,000$ SNPs). Data sets are generated using all pairwise combinations of:

$$S \in \{1, 2, 3, 4, 5\}, \quad (6)$$

$$\phi \in \{0.05, 0.10, 0.20, 0.40\}. \quad (7)$$

Two replicates are simulated for each of the 150 genes and each combination of (S, ϕ) , yielding a total of $2 \times 150 \times 5 \times 4 = 6\,000$ data sets.

Scenario 2 ($p = 3\,000$ – $12\,000$ SNPs). Data sets use the full set of SNPs near each gene (between 3 000 and 12 000), with $S = 10$ causal variants and $\phi = 0.3$. Two replicates per gene yield a total of $2 \times 150 = 300$ data sets.

1.1.4 SuSiE Model Hyperparameters Used in Simulations

When running SuSiE on the simulated data, the following settings are used:

- The maximum number of single-effect components is set to $L = 10$ for Scenario 1 and $L = 20$ for Scenario 2.
- The residual variance σ^2 is estimated from the data (via the ELBO-maximising empirical Bayes procedure described in the paper).
- The prior effect variance is fixed at $\sigma_{0l}^2 = 0.1 \cdot \text{var}(\mathbf{y})$ for all $l = 1, \dots, L$ (consistent with the real data application in Section 5 of the paper). The authors note that results are similar when σ_{0l}^2 is instead estimated from the data.
- The prior inclusion probabilities are set uniformly: $\boldsymbol{\pi} = (1/p, \dots, 1/p)$.

1.2 Post-Processing of Results

After fitting the SuSiE model via the IBSS algorithm, the raw output consists of L approximate posterior distributions for the single-effect vectors $\mathbf{b}_1, \dots, \mathbf{b}_L$, each parameterised by the triplet $(\boldsymbol{\alpha}_l, \boldsymbol{\mu}_{1l}, \boldsymbol{\sigma}_{1l})$ for $l = 1, \dots, L$. Two key quantities are derived from these: posterior inclusion probabilities (PIPs) and credible sets (CSs).

1.2.1 Posterior Inclusion Probabilities (PIPs)

Under the SuSiE variational approximation, the PIP for variable j is computed as:

$$\text{PIP}_j := \Pr(b^{(j)} \neq 0 \mid \mathbf{X}, \mathbf{y}) \approx 1 - \prod_{l \in \mathcal{L}} (1 - \alpha_{lj}), \quad (8)$$

where $b^{(j)} = \sum_{l=1}^L b_{lj}$ is the total effect of variable j , α_{lj} is the posterior probability that single-effect component l selects variable j , and $\mathcal{L} := \{l : \sigma_{0l}^2 > 0\}$ (to handle the case where some prior variances are estimated to be zero by the empirical Bayes procedure, effectively eliminating that component).

Note that this formula arises from the independence of the $\mathbf{b}_l$ under the variational approximation $q(\mathbf{b}_1, \dots, \mathbf{b}_L) = \prod_{l=1}^L q_l(\mathbf{b}_l)$: variable j has a non-zero total effect if and only if $b_{lj} \neq 0$ for at least one l .

1.2.2 Credible Set Construction

A level- ρ credible set is defined (Definition 1 in the paper) as a subset of variables that has probability ρ or greater of containing at least one effect variable. For each single-effect component $l = 1, \dots, L$, a credible set $\text{CS}(\alpha_l; \rho)$ is constructed as follows:

1. Sort the variables in decreasing order of α_{lj} .
2. Sequentially include variables in the set until their cumulative posterior probability exceeds ρ .

Formally, if π_l is a permutation such that $\alpha_{l,\pi_l(1)} \geq \alpha_{l,\pi_l(2)} \geq \dots \geq \alpha_{l,\pi_l(p)}$, then the level- ρ credible set for component l is:

$$\text{CS}(\alpha_l; \rho) = \{\pi_l(1), \dots, \pi_l(k^*)\}, \quad \text{where } k^* = \min \left\{ k : \sum_{i=1}^k \alpha_{l,\pi_l(i)} \geq \rho \right\}. \quad (9)$$

This yields L candidate credible sets in total. In all simulations and analyses, the level is set to $\rho = 0.95$.

1.2.3 Purity Filtering of Credible Sets

This is a critical post-processing step. When L exceeds the number of detectable effects in the data, many of the L credible sets will be large and contain weakly correlated variables (because the IBSS algorithm distributes posterior mass diffusely over many variables for the “excess” components). Such sets are uninformative and are filtered out.

Specifically, a *purity* measure is defined for each credible set as:

$$\text{purity}(\text{CS}) = \min_{j,k \in \text{CS}, j \neq k} |\text{cor}(x_j, x_k)|, \quad (10)$$

i.e., the minimum absolute pairwise correlation among all variables in the credible set. Any credible set with purity less than 0.5 is **discarded**. For credible sets containing more than 100 variables, the purity is approximated by sampling 100 variables at random from the set.

The authors note that this threshold of 0.5 was chosen primarily for comparability with DAP-G (Lee et al., 2018), which uses a similar threshold. In practice, the results are robust to this choice because most credible sets have purity either greater than 0.95 or less than 0.05, with intermediate cases being rare.

1.2.4 Determining the Number of Reported Credible Sets

The number of credible sets reported is *not* fixed *a priori*. Although L is specified as a (generous) upper bound on the number of effects, the actual number of reported credible sets is determined by how many survive the purity filter. In practice:

- Components whose prior variance σ_{0l}^2 is estimated to be zero (or near zero) by the empirical Bayes procedure effectively contribute no credible set.
- Components whose credible sets fail the purity threshold (< 0.5) are discarded.
- The remaining credible sets are reported.

This mechanism means that setting L larger than the true number of causal effects is generally safe: the excess components produce diffuse, low-purity credible sets that are automatically filtered out.

1.2.5 Summary of the Pipeline

1. Run IBSS (Algorithm 1) with specified L , estimating σ^2 and (optionally) σ_{0l}^2 .
2. Compute PIPs via Equation (3.12): $\text{PIP}_j \approx 1 - \prod_{l \in \mathcal{L}} (1 - \alpha_{lj})$.
3. For each $l \in \mathcal{L}$, construct a 95% credible set by sorting variables by α_{lj} and accumulating until the sum exceeds 0.95.
4. Discard any credible set with purity < 0.5 .
5. Report the surviving credible sets and the PIPs.

No other post-processing (e.g., recalibration of PIPs, LD-based merging, or additional thresholding) is applied. The pipeline from model output to evaluation is straightforward.

1.3 Evaluation Metrics

The SuSiE paper evaluates performance using two classes of output: (i) PIPs, and (ii) credible sets. Different metrics are used for each.

1.3.1 Metrics Based on Posterior Inclusion Probabilities

Power vs. False Discovery Rate (FDR) Curves. For a given PIP threshold $t \in [0, 1]$, a variable is “selected” if $\text{PIP}_j \geq t$. A selected variable is a *true positive* (TP) if $b_j \neq 0$ in the simulation, and a *false positive* (FP) otherwise. Let TP, FP, FN, and TN denote the counts of true positives, false positives, false negatives, and true negatives respectively across all variables in all simulations. Then:

$$\text{FDR} = \frac{\text{FP}}{\text{TP} + \text{FP}}, \quad (11)$$

$$\text{Power} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (12)$$

The authors note that this FDR is technically the *false discovery proportion* (i.e., calculated globally over all simulations, not averaged per-simulation). By varying t from 0 to 1, a power-vs-FDR curve is traced out. The authors note this is equivalent to a precision–recall curve (since precision = $1 - \text{FDR}$ and recall = power), with the x -axis reversed. Results are combined across all data sets within a scenario.

Specific PIP thresholds of $t = 0.90$ and $t = 0.95$ are highlighted on the curves.

PIP Calibration. PIP calibration is assessed visually: for each method, the PIPs are binned, and the empirical frequency of true causal variants in each bin is plotted against the nominal PIP value. A well-calibrated method produces points near the identity line. Calibration plots are provided in Figure S1 of the supplementary material.

Pairwise PIP Comparison (Scatter Plots). PIPs from SuSiE are plotted against those from each competing method (CAVIAR, FINEMAP, DAP-G) in scatter plots, with true effect variables coloured differently from non-effect variables. These are stratified by the number of causal variants ($S = 1$, $S = 2$, $S = 3-5$). Correlation between SuSiE PIPs and other methods’ PIPs is reported per data set (ranging from 0.94 to 1.0), along with the proportion of PIPs differing by more than 0.1.

1.3.2 Metrics Based on Credible Sets

Four metrics are used to evaluate 95% credible sets, computed as the mean (or median, where stated) over all credible sets across all data sets:

Coverage. The empirical (frequentist) coverage of a set of credible sets is the proportion that contain at least one true effect variable:

$$\text{Coverage} = \frac{\#\{\text{CSs containing } \geq 1 \text{ true effect variable}\}}{\#\{\text{total CSs reported}\}}. \quad (13)$$

Since these are Bayesian 95% credible sets, 95% frequentist coverage is not guaranteed, but the authors check that it is approximately achieved (typically slightly below 0.95, and in most cases above 0.90). Error bars showing ± 2 standard errors are included.

Power. The power of the credible set procedure is defined as the proportion of simulated effect variables that are included in *any* reported credible set:

$$\text{Power} = \frac{\#\{\text{true effect variables captured by at least one CS}\}}{\#\{\text{total simulated effect variables}\}}. \quad (14)$$

This is averaged over all data sets within a scenario.

Size. The median number of variables included in a credible set:

$$\text{Size} = \text{median}_l(|\text{CS}_l|), \quad (15)$$

where the median is taken over all reported credible sets across all simulations.

Purity (Average Squared Correlation). For comparing with DAP-G, which reports average squared correlation, purity in the comparison figures is measured as:

$$\text{Average } r^2 = \frac{1}{|\text{CS}|^2} \sum_{j \in \text{CS}} \sum_{k \in \text{CS}} \text{cor}(x_j, x_k)^2, \quad (16)$$

which is the average squared correlation among all pairs of variables within the credible set (including self-pairs). This statistic is reported because it is the output provided by DAP-G, enabling direct comparison. Note that this differs from the *minimum absolute correlation* used for the purity filtering threshold.

1.3.3 Comparison with Hierarchical Testing

When comparing with the hierarchical testing method (`hierinf`), an additional set of metrics is reported in tabular form (Table 3 of the paper):

- Number of credible sets (or significant clusters) reported.
- Power, Coverage, Median Size, and Average r^2 (as defined above).

For `hierinf`, the family-wise error rate (FWER) control level is set to $\alpha = 0.1$, chosen because it produced empirical coverage close to 0.95 in the simulations.

1.3.4 Computational Time

Run times (in seconds) are reported as mean, minimum, and maximum across simulations with $S = 3$ causal variants (Table 2 of the paper), providing a benchmark for computational scalability.

2 BEATRICE

2.1 Simulation Design

Ghosal et al. simulate data using a mixed linear model that explicitly separates the contributions of causal variants (fixed effects) from noncausal variants (random effects), allowing controlled evaluation under varying degrees of polygenicity. Unlike many fine-mapping benchmarks that use real genotype matrices, BEATRICE uses *simulated* genotypes.

2.1.1 Genotype Simulation

Genotypes are simulated using the method of Dimitromanolakis et al. (2019), based on data from the 1000 Genomes Project. Specifically:

- An arbitrary sub-region of approximately 1 Mb (39.9 Mb–40.9 Mb) from Chromosome 2 is selected as the base haplotype panel.
- Rare variants with minor allele frequency (MAF) < 0.02 are removed, leaving approximately 3 500 variants.
- Pairs of haplotypes are simulated to generate 10 000 unrelated individuals.
- No further filtering of variants is applied to the simulated data.
- For each simulation replicate, $m = 1\,000$ variants and $n = 5\,000$ individuals are randomly selected to form the genotype matrix $\mathbf{X} \in \mathbb{R}^{n \times m}$.
- The columns of $\mathbf{X}$ are normalised to have mean 0 and variance 1, i.e. $\frac{1}{n} \sum_i X_{ij} = 0$ and $\frac{1}{n} \sum_i X_{ij}^2 = 1$ for $j = 1, \dots, m$.

2.1.2 Generative Model for the Phenotype

The phenotype vector $\tilde{\mathbf{y}} \in \mathbb{R}^n$ is generated from a mixed linear model that decomposes the genetic contribution into causal (fixed) and noncausal (random) effects. Let C denote the set of d randomly selected causal variants, $\mathbf{X}_C \in \mathbb{R}^{n \times d}$ the corresponding genotype sub-matrix, and $\mathbf{X}_{NC} \in \mathbb{R}^{n \times (m-d)}$ the remaining noncausal variants. The raw phenotype is generated as:

$$\mathbf{y} = \underbrace{\mathbf{X}_C \boldsymbol{\beta}}_{\mathbf{g}_C} + \underbrace{\mathbf{g}_{NC}}_{\text{noncausal}} + \boldsymbol{\epsilon}, \quad (17)$$

where:

$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d), \quad (18)$$

$$\mathbf{g}_{NC} \sim \mathcal{N}\left(\mathbf{0}, \frac{1}{m-d} \mathbf{X}_{NC} \mathbf{X}_{NC}^T\right), \quad (19)$$

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \alpha^2 \mathbf{I}_n). \quad (20)$$

The key distinction from many other fine-mapping simulations is the inclusion of the term $\mathbf{g}_{NC}$, which models infinitesimal polygenic effects from noncausal variants. This creates a more challenging scenario where noncausal variants can have inflated GWAS statistics due to both LD with causal variants *and* their own small direct effects.

2.1.3 Variance Normalisation

Two parameters control the signal-to-noise structure:

- ω^2 : the total phenotypic variance explained by genetics (both causal and noncausal);

- p : the proportion of the genetic variance attributable to the causal variants.

The phenotype is normalised following the strategy of Meyer and Birney (2018):

$$\tilde{\mathbf{y}} = \sqrt{\frac{p \cdot \omega^2}{\text{var}(\mathbf{g}_C)}} \mathbf{g}_C + \sqrt{\frac{(1-p) \cdot \omega^2}{\text{var}(\mathbf{g}_{NC})}} \mathbf{g}_{NC} + \tilde{\boldsymbol{\epsilon}}, \quad \tilde{\boldsymbol{\epsilon}} \sim \mathcal{N}(\mathbf{0}, (1-\omega^2)\mathbf{I}_n), \quad (21)$$

where $\text{var}(\mathbf{g}_C)$ and $\text{var}(\mathbf{g}_{NC})$ are the empirical (sample) variances of the vectors $\mathbf{g}_C$ and $\mathbf{g}_{NC}$, respectively. After normalisation, the causal variants explain a fraction $p \cdot \omega^2$ of the total phenotypic variance and the noncausal variants explain $(1-p) \cdot \omega^2$.

2.1.4 Computing Summary Statistics

The normalised phenotype $\tilde{\mathbf{y}}$ is treated as the input to a standard GWAS pipeline. For each variant i , the marginal effect size $\hat{\beta}_i$ is estimated via ordinary least squares regression of $\tilde{\mathbf{y}}$ on the *unnormalised* genotype column $\tilde{x}_i$. The effect sizes are then converted to z -scores:

$$z_i = \frac{\hat{\beta}_i}{\text{se}(\hat{\beta}_i)}, \quad (22)$$

where $\text{se}(\cdot)$ denotes the standard error. The LD matrix is computed from the *normalised* genotype data as $\boldsymbol{\Sigma}_X = \frac{1}{n} \mathbf{X}^T \mathbf{X}$. The z -scores $\mathbf{z}$ and LD matrix $\boldsymbol{\Sigma}_X$ are the inputs to all fine-mapping methods.

2.1.5 Simulation Parameters

The following parameter ranges are swept:

$$d \in \{1, 4, 8, 12\} \quad (\text{number of causal variants}), \quad (23)$$

$$\omega^2 \in \{0.1, 0.2, 0.4, 0.5, 0.7, 0.8\} \quad (\text{total genotype variance}), \quad (24)$$

$$p \in \{0.1, 0.3, 0.5, 0.7, 0.9\} \quad (\text{proportion of genetic variance from causal variants}). \quad (25)$$

For each parameter combination, 20 independent replicates are generated by re-sampling causal variant locations, effect sizes $\boldsymbol{\beta}$, the noncausal component $\mathbf{g}_{NC}$, and the noise $\tilde{\boldsymbol{\epsilon}}$. This yields a total of $4 \times 6 \times 5 \times 20 = 2400$ simulated data sets.

Note for benchmarking: The inclusion of a noncausal random-effect component $\mathbf{g}_{NC}$ is non-standard compared to the purely sparse simulations used by, e.g., SuSiE and FINEMAP. When p is small (e.g., $p = 0.1$), only 10% of the genetic variance comes from the causal variants, making the problem substantially harder. This design choice favours methods that can accommodate infinitesimal effects, and the resulting metrics may not be directly comparable with benchmarks that assume a purely sparse generative model.

2.2 Post-Processing of Results

BEATRICE uses a multi-step post-processing pipeline that differs substantially from the cumulative-PIP credible set construction used by SuSiE and FINEMAP.

2.2.1 From Optimisation to Causal Configurations

During training, the BEATRICE optimisation procedure (Algorithm 1 in the paper) generates, at each gradient descent iteration t , a sample causal vector $\mathbf{c}_t^l \in [0, 1]^m$ from the binary concrete proposal distribution $q(\cdot; \mathbf{p}(\phi_t), \lambda)$ via:

$$c_i = \xi \left(\frac{\log\left(\frac{U}{1-U}\right) + \log\left(\frac{p_i}{1-p_i}\right)}{\lambda} \right), \quad U \sim \text{Uniform}(0, 1), \quad (26)$$

where $\xi(\cdot)$ is the sigmoid function and λ is the temperature parameter (annealed to 0.01).

Each continuous sample $\mathbf{c}_t^l$ is then **binarised** to a vector $\mathbf{b}_t^l \in \{0, 1\}^m$ via:

$$b_i^l = \begin{cases} 1 & \text{if } c_i^l > \gamma, \\ 0 & \text{otherwise,} \end{cases} \quad (27)$$

where $\gamma = 0.1$ is the default binarisation threshold. The binarised vectors are accumulated into a **reduced set of causal configurations** $\mathcal{B}_R$ over the course of optimisation.

Sparsity pruning. Before computing the loss at each iteration, the set $S_t^l = \{i : c_i^l > 0.01\}$ is pruned to contain at most 50 indices. All entries of $\mathbf{c}_t^l$ outside this pruned set are set to zero. This limits computational cost and enforces sparsity.

2.2.2 Computing Posterior Inclusion Probabilities

After optimisation, the posterior probability of each binarised causal configuration $\mathbf{b} \in \mathcal{B}_R$ is approximated using Bayes' rule over the *reduced* set:

$$p(\mathbf{b} \mid \mathbf{z}, \Sigma_X) = \frac{p(\mathbf{z} \mid \Sigma_X, \mathbf{b}) p(\mathbf{b})}{\sum_{\mathbf{b}' \in \mathcal{B}_R} p(\mathbf{z} \mid \Sigma_X, \mathbf{b}') p(\mathbf{b}')}, \quad (28)$$

where the likelihood $p(\mathbf{z} \mid \Sigma_X, \mathbf{b})$ is evaluated using the multivariate normal from Equation (4) of the paper, substituting the binarised configuration $\mathbf{b}$ for the diagonal of Σ_C . Note that the denominator sums over $\mathcal{B}_R$ rather than the full exponential space of 2^m configurations.

The PIP for each variant i is then computed by marginalising over all configurations in $\mathcal{B}_R$ that include variant i :

$$\text{PIP}_i = P(b_i = 1 \mid \mathbf{z}, \Sigma_X) \approx \sum_{\mathbf{b} \in \mathcal{S}} p(\mathbf{b} \mid \mathbf{z}, \Sigma_X), \quad \mathcal{S} = \{\mathbf{b} \in \mathcal{B}_R : b_i = 1\}. \quad (29)$$

Note for benchmarking: The quality of the PIP estimates depends entirely on the richness of $\mathcal{B}_R$. If the optimisation does not sufficiently explore the configuration space, the reduced set may miss important configurations, and the PIPs will be biased. The binarisation threshold $\gamma = 0.1$ also introduces a hard cutoff: any variant whose continuous causal probability c_i never exceeds 0.1 during training will never appear in $\mathcal{B}_R$ and will have $\text{PIP} = 0$.

2.2.3 Credible Set Construction

BEATRICE constructs credible sets using a two-step conditional procedure described in Supplementary Section S1.3 of the paper:

Step 1: Identify “key” variants (index variants). Using a conditional stepwise procedure, the variant with the highest *conditional* posterior probability (given previously selected variants) is iteratively added to a set of key variants. This continues until no further variant has a sufficiently high conditional probability. The number of key variants determines the number of credible sets.

Step 2: Build credible sets around each key variant. For each key variant, the conditional inclusion probabilities of all other variants (given the key variants) are computed. Variants are added to the credible set in decreasing order of their conditional inclusion probability until the cumulative probability reaches the desired coverage level (0.95).

Note for benchmarking: This conditional stepwise credible set procedure is *fundamentally different* from the cumulative-PIP approach used by SuSiE, where credible sets arise naturally

from the single-effect decomposition. BEATRICE’s approach is more similar to the signal-cluster heuristics of DAP-G. The conditional construction means that the number and composition of credible sets depends on the order in which key variants are selected, introducing a sequential dependence not present in SuSiE’s parallel decomposition.

2.2.4 No Purity Filtering

Unlike SuSiE, BEATRICE does **not** apply a purity filter (minimum absolute correlation threshold) to discard uninformative credible sets. All credible sets identified by the conditional procedure are reported. This means that BEATRICE may report credible sets containing weakly correlated variants, which in SuSiE would be filtered out by the purity < 0.5 threshold.

2.2.5 Summary of the Pipeline

1. Run the neural-network-based variational optimisation (Algorithm 1), collecting binarised causal configurations $\mathcal{B}_R$ during training.
2. Compute posterior configuration probabilities over $\mathcal{B}_R$ via Bayes’ rule.
3. Compute PIPs by marginalising over $\mathcal{B}_R$.
4. Identify key variants via conditional stepwise selection.
5. Construct credible sets around each key variant using conditional inclusion probabilities at level $\rho = 0.95$.

2.3 Evaluation Metrics

BEATRICE evaluates performance using two classes of output: PIPs and credible sets. The metrics differ from those used by SuSiE in several important respects.

2.3.1 Metrics Based on Posterior Inclusion Probabilities

AUPRC (Area Under the Precision–Recall Curve). The primary PIP-based metric is the **AUPRC**. A threshold t is swept across the range of PIP values, and at each threshold, variants with $\text{PIP} \geq t$ are classified as “predicted causal.” For each threshold:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad (30)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad (31)$$

where TP, FP, and FN denote true positives, false positives, and false negatives, respectively, against the true binary causal status. The precision–recall curve is plotted and the area underneath is computed.

AUPRC is preferred over AUROC (area under the ROC curve) because it is more informative under severe class imbalance—which is the case in fine mapping, where the number of causal variants (d) is much smaller than m . BEATRICE reports mean AUPRC and 95% confidence intervals computed across simulation configurations.

Comparison with SuSiE’s FDR–Power curve: Note that SuSiE reports Power vs. FDR ($= 1 - \text{Precision}$) curves, which are mathematically equivalent to precision–recall curves with a transformed x -axis. However, SuSiE computes these *globally* across all simulations, while BEATRICE reports the mean across individual simulation configurations. The aggregation strategy may affect the resulting curves, especially when there are many simulations with different numbers of causal variants.

Power vs. FDR Curves (Supplementary). In Supplementary Section S1.8, BEATRICE additionally provides Power vs. FDR plots for different PIP thresholds, matching the format used by SuSiE. Here:

$$\text{Power (Recall)} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad (32)$$

$$\text{FDR} = \frac{\text{FP}}{\text{TP} + \text{FP}}. \quad (33)$$

These are computed by sweeping a PIP threshold and are identical to the precision–recall formulation above.

2.3.2 Metrics Based on Credible Sets

Three metrics are reported for credible sets, following the definitions of Zou et al. (2022) and Wang et al. (2020):

Coverage. The proportion of credible sets that contain at least one true causal variant:

$$\text{Coverage} = \frac{\#\{\text{CSs containing } \geq 1 \text{ true causal variant}\}}{\#\{\text{total CSs reported}\}}. \quad (34)$$

Power. The proportion of true causal variants captured by *any* reported credible set:

$$\text{Power} = \frac{\#\{\text{true causal variants in at least one CS}\}}{\#\{\text{total true causal variants}\}}. \quad (35)$$

Size. The average number of variants in a credible set (reported as a mean, not a median as in SuSiE).

2.3.3 How Results Are Aggregated

Results are presented in three ways, each fixing one simulation parameter and averaging over the others:

1. **Varying d** (number of causal variants): fix $d = d^*$, compute mean and 95% CI across all (ω^2, p) combinations.
2. **Varying ω^2** (total genotype variance): fix $\omega^2 = \omega^{*2}$, compute mean and 95% CI across all (d, p) combinations.
3. **Varying p** (causal proportion): fix $p = p^*$, compute mean and 95% CI across all (d, ω^2) combinations.

In each case, the mean is computed over $20 \times |\text{remaining parameters}|$ replicates. Error bars represent 95% confidence intervals.

2.3.4 Differences from SuSiE’s Evaluation

Several methodological differences should be noted when comparing BEATRICE’s evaluation with that of SuSiE:

1. **Generative model.** BEATRICE simulations include a noncausal random-effect component ($\mathbf{g}_{NC}$); SuSiE simulations use a purely sparse model with no infinitesimal effects. This makes direct metric comparison across papers problematic.

2. **Genotype source.** BEATRICE uses simulated genotypes (from 1000 Genomes haplotypes); SuSiE uses real GTEx genotype matrices.
3. **Primary PIP metric.** BEATRICE uses AUPRC as its primary summary; SuSiE uses power-vs-FDR curves with highlighted PIP thresholds.
4. **Credible set size.** BEATRICE reports *mean* size; SuSiE reports *median* size.
5. **No purity filtering.** BEATRICE does not filter credible sets by purity, while SuSiE discards sets with minimum absolute correlation < 0.5 .
6. **No purity metric.** BEATRICE does not report a purity metric for credible sets.
7. **PIP calibration.** BEATRICE does not provide PIP calibration plots.

3 FINEMAP

3.1 Simulation Design

Benner et al. simulate data using *real* genotype data from a Finnish population cohort combined with synthetic phenotype vectors. The primary goal of the simulations is to benchmark computational efficiency and accuracy of the shotgun stochastic search (SSS) algorithm against the exhaustive search of CAVIARBF, rather than to provide a comprehensive evaluation of fine-mapping performance per se.

3.1.1 Genotype Data (Design Matrix $\mathbf{X}$)

Real genotype data are obtained from $n = 18\,834$ individuals in the National FINRISK study. The data comprise a 500 kilobase region centred on rs11591147 in the *PCSK9* gene on chromosome 1, containing 1920 polymorphic SNPs with pairwise absolute correlations less than 0.99 (i.e., near-perfectly correlated SNP pairs are removed).

3.1.2 Generative Model for the Phenotype

The phenotype vector $\mathbf{y} \in \mathbb{R}^n$ is generated under the linear model:

$$\mathbf{y} = \sum_{c \in C} \beta_c \mathbf{g}_c + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n), \quad (36)$$

where C is the set of causal SNPs, $\mathbf{g}_c$ is the genotype vector at the c -th causal SNP, and β_c and f_c are, respectively, the effect size and minor allele frequency of the c -th causal SNP. The residual variance is set to:

$$\sigma^2 = 1 - \sum_{c \in C} 2f_c(1 - f_c)\beta_c^2, \quad (37)$$

which ensures that the total phenotypic variance is normalised to 1 (assuming the genotypes contribute variance $\sum_c 2f_c(1 - f_c)\beta_c^2$ under Hardy–Weinberg equilibrium).

3.1.3 Causal Variant Selection and Effect Sizes

In both simulation scenarios, the true number of causal SNPs is **five**. Causal SNPs are randomly chosen under a specific constraint: each causal SNP must have at least one highly correlated proxy (absolute correlation > 0.5) among the other (non-causal) variants. This design choice creates a challenging LD structure that stress-tests the search algorithm’s ability to distinguish true causal variants from their correlated proxies.

Effect sizes β_c are calibrated so that the statistical power at a genome-wide significance level of 5×10^{-8} is approximately 0.5 for each causal SNP. The paper does not specify the exact

distribution from which effect sizes are drawn—rather, they are deterministically set to achieve the target power.

3.1.4 Computing Summary Statistics

Single-SNP association testing is performed using a linear model to compute marginal z -scores. The SNP correlation matrix $\mathbf{R} = n^{-1}\mathbf{X}^T\mathbf{X}$ is computed from the same genotype data. The z -scores and correlation matrix are the inputs to both FINEMAP and CAVIARBF.

3.1.5 Simulation Scenarios

Two scenarios are considered:

Scenario A (varying m , fixed K). The number of SNPs is varied: $m \in \{750, 1000, 1250, 1500\}$, with causal configurations of up to $K = 3$ or $K = 5$ SNPs. Five hundred data sets are generated per combination of m and K .

Scenario B (fixed m , varying K). A fixed number of $m = 150$ SNPs is used, with the maximum number of causal SNPs varied: $K \in \{1, 2, 3, 4, 5\}$. Five hundred data sets are generated per value of K .

3.1.6 FINEMAP Hyperparameters Used in Simulations

- The prior standard deviation for causal effects is set to $s_\lambda = 0.05$ (equivalently, $s_\lambda^2 = 0.0025$), meaning that with 95% probability a causal SNP explains less than 1% of trait variation.
- The prior on the number of causal SNPs uses the distribution:

$$p(\gamma) \propto \binom{m}{k}^{-1} \left(\frac{1}{m}\right)^k \left(\frac{m-1}{m}\right)^{m-k}, \quad \text{for } k = 1, \dots, K, \quad (38)$$

which corresponds to each SNP being independently causal with probability $1/m$, conditioned on at least one causal SNP ($p_0 = 0$).

- FINEMAP is run with 100 SSS iterations, saving the top 50 000 evaluated causal configurations.

Note for benchmarking: The FINEMAP simulations differ substantially from those of SuSiE and BEATRICE. The phenotype model is purely sparse (no infinitesimal polygenic effects), uses a single real genotype region (not multiple genes), and calibrates effect sizes to a fixed power target rather than drawing them from a distribution. The primary comparison is FINEMAP vs. CAVIARBF (accuracy of stochastic vs. exhaustive search), not a broad fine-mapping benchmark.

3.2 Post-Processing of Results

FINEMAP’s post-processing is conceptually straightforward: posterior probabilities over causal configurations are computed from the stochastic search output, and all downstream quantities are derived by marginalisation.

3.2.1 Posterior Probabilities of Causal Configurations

The SSS algorithm explores the space of causal configurations and stores each visited configuration γ along with its unnormalised posterior probability $p^*(\gamma \mid \mathbf{y}, \mathbf{X})$ in a list Γ^* . The normalised

posterior is computed over this stored set:

$$p(\boldsymbol{\gamma} \mid \mathbf{y}, \mathbf{X}) = \frac{p^*(\boldsymbol{\gamma} \mid \mathbf{y}, \mathbf{X})}{\sum_{\boldsymbol{\gamma}' \in \Gamma^*} p^*(\boldsymbol{\gamma}' \mid \mathbf{y}, \mathbf{X})}. \quad (39)$$

The unnormalised posterior combines the prior and the marginal likelihood (Bayes factor):

$$p^*(\boldsymbol{\gamma} \mid \mathbf{y}, \mathbf{X}) = \binom{m}{k}^{-1} p_k \times \text{BF}(\boldsymbol{\gamma} : \text{NULL}), \quad (40)$$

where $k = \sum_{\ell} \gamma_{\ell}$ and p_k is the prior probability that there are exactly k causal SNPs.

3.2.2 Posterior Inclusion Probabilities (Single-SNP Inclusion Probabilities)

The marginal posterior probability that the ℓ -th SNP is causal—termed the “single-SNP inclusion probability” by the FINEMAP authors—is computed by summing over all configurations in Γ^* that include SNP ℓ :

$$p(\gamma_{\ell} = 1 \mid \mathbf{y}, \mathbf{X}) = \sum_{\boldsymbol{\gamma} \in \Gamma^*} \mathbf{1}(\gamma_{\ell} = 1) p(\boldsymbol{\gamma} \mid \mathbf{y}, \mathbf{X}). \quad (41)$$

This is equivalent to the PIP as defined by other methods, though normalised only over the explored set Γ^* rather than the full configuration space.

3.2.3 Single-SNP Bayes Factors

FINEMAP additionally computes a single-SNP Bayes factor for each variant:

$$\text{BF}(\gamma_{\ell} = 1 : \gamma_{\ell} = 0) = \frac{p(\gamma_{\ell} = 1 \mid \mathbf{y}, \mathbf{X}) / p(\gamma_{\ell} = 0 \mid \mathbf{y}, \mathbf{X})}{p(\gamma_{\ell} = 1) / p(\gamma_{\ell} = 0)}, \quad (42)$$

where the prior inclusion probability is $p(\gamma_{\ell} = 1) = \sum_{k=1}^K (k/m) p_k$.

3.2.4 Credible Sets

The original FINEMAP paper (2016) does **not** describe a credible set construction procedure. The paper references the concept of “credible sets of causal variants” from Maller et al. (2012), but does not implement or evaluate credible sets in the simulation study. Later versions of the FINEMAP software (post-v1.0) do output credible sets, but the methodology for their construction is not detailed in this paper.

Note for benchmarking: When credible sets from FINEMAP appear in other papers’ benchmarks (e.g., SuSiE, SuSiE-RSS), they are typically constructed post hoc by sorting variants within each detected signal by their conditional inclusion probability and accumulating until a 95% threshold is reached—similar to the SuSiE approach but applied to FINEMAP’s output. The exact procedure depends on the FINEMAP software version used and is not standardised in the original publication.

3.2.5 Handling of Collinear SNPs

FINEMAP implements an option to set the posterior probability of a causal configuration to zero if it contains at least one pair of SNPs with absolute correlation exceeding a user-specified threshold. This prevents numerical instability when the correlation sub-matrix $\mathbf{R}_{CC}$ is near-singular.

3.2.6 Summary of the Pipeline

1. Run the SSS algorithm for a fixed number of iterations (default: 100), exploring causal configurations by adding, deleting, or swapping causal SNPs.
2. Store all visited configurations and their unnormalised posterior probabilities in Γ^* (saving the top 50 000 by default).
3. Normalise posterior probabilities over Γ^* .
4. Compute PIPs (single-SNP inclusion probabilities) by marginalisation over Γ^* .
5. Compute single-SNP Bayes factors.
6. Output the ranked list of causal configurations and per-SNP summaries.

No purity filtering, credible set construction, or additional post-processing is described in the original paper.

3.3 Evaluation Metrics

The FINEMAP paper focuses primarily on two aspects: *computational efficiency* (speed comparison with exhaustive search) and *accuracy of the stochastic search* (agreement with exhaustive search results). It does **not** use the standard credible-set-based metrics (coverage, power, size) that are used by SuSiE and BEATRICE.

3.3.1 Accuracy of Single-SNP Inclusion Probabilities

The primary accuracy metric is the **maximum absolute difference** between FINEMAP’s and CAVIARBF’s single-SNP inclusion probabilities (PIPs), computed per data set:

$$\Delta_{\max} = \max_{\ell=1,\dots,m} |\text{PIP}_{\ell}^{\text{FINEMAP}} - \text{PIP}_{\ell}^{\text{CAVIARBF}}|. \quad (43)$$

This is reported as percentiles (median, 95th, 99th, and maximum) across the 500 simulated data sets in Scenario B. The rationale is that CAVIARBF performs exact exhaustive enumeration (up to the maximum K), so agreement with CAVIARBF demonstrates that the SSS algorithm is not missing important configurations.

3.3.2 Proportion of Causal SNPs Recovered (Fine-Mapping Accuracy Curve)

For Scenario A ($m = 1\,500$ SNPs, 5 true causal SNPs), the authors evaluate fine-mapping accuracy by the following procedure:

1. Rank all m SNPs by their single-SNP inclusion probabilities (PIPs) in decreasing order.
2. For each value of $T = 1, 2, \dots$, compute the proportion of the 5 true causal SNPs that appear among the top T ranked SNPs.

This produces a curve of “proportion of causal SNPs included” vs. “number of top SNPs selected,” averaged over 500 data sets. This is plotted for FINEMAP with $K = 3$ and $K = 5$, and CAVIARBF with $K = 3$ (CAVIARBF cannot handle $K = 5$ at $m = 1\,500$). A method that assigns higher PIPs to true causal SNPs will reach a proportion of 1.0 with fewer top SNPs.

Comparison with other methods’ metrics: This curve is *not* the same as a power-vs-FDR curve or a precision–recall curve. It does not threshold on PIP values; instead, it thresholds on rank. It is closer to a “recall at top- T ” curve. No FDR or precision is computed.

3.3.3 Computational Efficiency

Processing time (in seconds) is reported on a $\log_{10}$ scale for both FINEMAP and CAVIARBF, averaged over 500 data sets, using a single core of an Intel Haswell E5-2690v3 processor at 2.6 GHz. Times are reported for:

- Scenario A: increasing m at fixed $K \in \{3, 5\}$.
- Scenario B: increasing K at fixed $m = 150$.

3.3.4 Metrics *Not* Reported

The following metrics, which are standard in later fine-mapping benchmarks, are **not reported** in the original FINEMAP paper:

- Credible set coverage, power, size, or purity.
- Power vs. FDR (or precision–recall) curves.
- PIP calibration plots.
- AUPRC or AUROC.

3.3.5 Differences from SuSiE’s and BEATRICE’s Evaluation

1. **Benchmark target.** FINEMAP benchmarks against CAVIARBF’s exhaustive search (is the stochastic search accurate?), not against ground truth causal status (is the model good at fine-mapping?). SuSiE and BEATRICE benchmark against ground truth.
2. **No credible sets.** The original paper does not evaluate credible sets. Credible set comparisons involving FINEMAP in other papers (e.g., SuSiE-RSS, Zou et al. 2022) use later software versions.
3. **Simulation scale.** FINEMAP uses a single genomic region from one cohort (500 datasets per scenario); SuSiE uses 150 gene regions (6 000 data sets); BEATRICE uses simulated genotypes (2 400 data sets).
4. **No FDR/power curves.** The primary accuracy metric is agreement with exhaustive search, not classification of causal vs. non-causal variants.
5. **Effect size calibration.** FINEMAP calibrates effect sizes to achieve approximately 50% power at 5×10^{-8} ; SuSiE draws effect sizes from $\mathcal{N}(0, 0.6^2)$ and varies PVE; BEATRICE draws from $\mathcal{N}(0, \mathbf{I}_d)$ and varies variance components.

4 PAINTOR

4.1 Simulation Design

Kichaev et al. simulate fine-mapping data across multiple risk loci simultaneously, using genotypes derived from 1000 Genomes haplotypes. A distinguishing feature of PAINTOR’s simulation framework is the explicit modelling of functional annotation enrichment: causal variants are preferentially drawn from synthetic functional categories.

4.1.1 Genotype Simulation

Genotypes are simulated using HAPGEN, starting from 1000 Genomes (1KG) European reference haplotypes. For each simulation:

- One hundred 10 Kb loci are randomly selected.

- Monomorphic and rare variants ($\text{MAF} < 0.01$) are filtered out.
- Genotypes are normalised to be mean-centred with unit variance.
- The sample size is $N = 10\,000$ individuals.

For simulations requiring loci larger than 10 Kb (25 Kb and 50 Kb variants are also tested), z -scores are instead drawn directly from a multivariate normal distribution with covariance equal to the LD matrix from the European 1KG panel and non-centrality parameters at causal sites drawn from $\mathcal{N}(5, 0.2^2)$.

4.1.2 Functional Annotations

Three synthetic binary functional annotations are created, roughly corresponding to:

- Coding exons: 2.2% of all variants;
- Transcription start sites (TSS): 2.2% of all variants;
- DNase hypersensitivity sites (DHS): 30.7% of all variants.

SNPs are randomly assigned to these annotations at the above fixed proportions.

4.1.3 Causal Variant Selection

Causal status for each SNP is drawn according to a logistic model that links annotation membership to causal probability:

$$P(c_{ij} = 1) = \frac{\exp(\mathbf{a}_{ij}^T \boldsymbol{\gamma})}{1 + \exp(\mathbf{a}_{ij}^T \boldsymbol{\gamma})}, \quad (44)$$

where $\mathbf{a}_{ij}$ is the annotation vector for SNP i at locus j (with a leading 1 for the baseline intercept), and $\boldsymbol{\gamma}$ is the vector of annotation effect sizes. The parameters $\boldsymbol{\gamma}$ are set to induce the desired enrichment levels:

- Coding exons: ~ 9.5 -fold enrichment of causal variants;
- TSS: ~ 5.7 -fold enrichment;
- DHS: ~ 3.65 -fold enrichment.

The $\boldsymbol{\gamma}$ values are calibrated to produce approximately one causal variant per locus in expectation. This random assignment results in loci with zero (~ 36), one (~ 34), or multiple (~ 30) causal variants on average.

4.1.4 Generative Model for the Phenotype

A continuous phenotype is simulated using a linear model. The causal SNPs collectively explain a fixed proportion of phenotypic variance $h^2 = 0.25$. This variance is partitioned *equally* among all causal SNPs. Specifically, the m -th individual's phenotype is drawn as:

$$y_m = \sum_{k=1}^K \beta_k g_{mk} + \epsilon_m, \quad \epsilon_m \sim \mathcal{N}(0, 1 - h^2), \quad (45)$$

where K is the total number of causal variants across all 100 loci, β_k is the effect size of the k -th causal SNP (set so that each explains h^2/K of the phenotypic variance), and g_{mk} is the number of copies of the risk allele (randomly assigned as reference or alternate) for individual m . The authors note that qualitatively similar results are obtained when variance is unevenly partitioned among causal variants (drawing effect sizes from $\mathcal{N}(0, 1)$ and renormalising).

4.1.5 Computing Summary Statistics

Association z -scores are computed at each SNP by taking the Wald statistic from marginal regression of the phenotype onto each SNP’s genotype. The LD matrix Σ is computed from the genotype data (or from the 1KG reference panel for larger loci). These z -scores and the LD matrix are the inputs to PAINTOR and competing methods.

4.1.6 Simulation Parameters

The default simulation uses:

- 100 loci of 10 Kb each;
- $N = 10\,000$ individuals;
- $h^2 = 0.25$ (total heritability across all loci);
- Three functional annotations with enrichments of $9.5\times$, $5.7\times$, and $3.65\times$;
- 500 independent simulation replicates.

Additional simulations vary locus size ($\{10, 25, 50\}$ Kb), sample size, and annotation enrichment levels (from 8-fold depletion to 8-fold enrichment).

Note for benchmarking: The maximum number of causal variants per locus that PAINTOR integrates over is restricted to ≤ 3 for computational tractability (exhaustive enumeration of all $\binom{m}{k}$ configurations for $k \leq 3$). This is a hard constraint of the method. Furthermore, the simulation framework assigns causal status via the logistic annotation model—the same model PAINTOR uses for inference—which may favour PAINTOR relative to methods that do not use functional annotations.

4.2 Post-Processing of Results

PAINTOR’s post-processing pipeline produces per-variant posterior probabilities of causality and ρ -level credible sets. Unlike SuSiE, PAINTOR does not decompose the posterior into single-effect components; instead, it marginalises over all enumerated causal configurations.

4.2.1 Posterior Probabilities of Causality

For each locus j , the posterior probability of a particular causal configuration $\mathbf{c}_j$ is computed via Bayes’ rule:

$$P(\mathbf{c}_j \mid \mathbf{z}_j, \Sigma_j) = \frac{P(\mathbf{z}_j \mid \mathbf{c}_j, \Sigma_j) P(\mathbf{c}_j \mid \gamma)}{\sum_{\mathbf{c}'_j} P(\mathbf{z}_j \mid \mathbf{c}'_j, \Sigma_j) P(\mathbf{c}'_j \mid \gamma)}, \quad (46)$$

where the sum in the denominator runs over *all* causal configurations with ≤ 3 causal variants (exhaustive enumeration). The likelihood term uses a multivariate normal model:

$$P(\mathbf{z}_j \mid \mathbf{c}_j, \Sigma_j) = \mathcal{N}(\mathbf{z}_j \mid \Sigma_j(\boldsymbol{\lambda}_j \circ \mathbf{c}_j), \Sigma_j), \quad (47)$$

where $\boldsymbol{\lambda}_j$ is the vector of non-centrality parameters (NCPs) and $\circ$ denotes elementwise multiplication. The NCPs are **fixed** (not integrated out): they are set to the observed z -score if $|z| > 3.7$, or to $\text{sign}(z) \times 3.7$ otherwise.

The prior probability of each configuration is determined by the logistic annotation model, with parameters γ estimated via EM across all loci.

4.2.2 Per-Variant Posterior Inclusion Probabilities

The PIP for SNP i at locus j is obtained by marginalising over all configurations in which that SNP is causal:

$$\text{PIP}_{ij} = \sum_{\mathbf{c}_j: c_{ij}=1} P(\mathbf{c}_j \mid \mathbf{z}_j, \Sigma_j). \quad (48)$$

Since the enumeration is exhaustive (up to 3 causals), these PIPs are exact under the model assumptions.

4.2.3 Credible Set Construction

PAINTOR defines an ρ -level “causal set” (credible set) as the set of top SNPs, ranked by their posterior probability, *across all fine-mapping loci*, that consume an ρ fraction of the *total* posterior probability mass. Formally:

1. Rank all SNPs across all loci by their PIPs in decreasing order.
2. Sequentially include SNPs until the cumulative sum of PIPs reaches ρ (e.g., 0.90 or 0.95) of the total posterior mass.

Note for benchmarking: This is a *global* credible set across all loci, not a per-locus or per-signal credible set as in SuSiE. The “90% confidence set” in PAINTOR means the smallest set of SNPs across all loci whose PIPs sum to 90% of the total posterior mass. This is fundamentally different from SuSiE’s per-signal credible sets with 95% coverage probability. The two definitions are not directly comparable. No purity filtering is applied.

4.2.4 Non-Centrality Parameter Handling

A notable post-processing detail: PAINTOR fixes the NCPs rather than integrating them out (as done by FINEMAP and CAVIARBF). This means the likelihood evaluation depends on a point estimate of the NCP, introducing a potential source of bias. The authors set:

$$\lambda_{ij} = \begin{cases} z_{ij} & \text{if } |z_{ij}| > 3.7, \\ \text{sign}(z_{ij}) \times 3.7 & \text{otherwise.} \end{cases} \quad (49)$$

This choice was validated via simulations comparing several NCP approximation strategies.

4.2.5 Summary of the Pipeline

1. Estimate annotation effect parameters γ via EM across all loci.
2. For each locus, exhaustively enumerate all causal configurations with ≤ 3 causal variants.
3. Compute posterior configuration probabilities using the annotation-informed prior and the multivariate normal likelihood.
4. Compute per-variant PIPs by marginalisation.
5. Rank all variants across all loci by PIP and form global ρ -level credible sets.

4.3 Evaluation Metrics

PAINTOR’s evaluation framework differs significantly from both SuSiE and FINEMAP. The primary metric is a “proportion of causal variants identified” curve as a function of the number of SNPs selected for follow-up, reflecting the paper’s framing of fine-mapping as a variant prioritisation problem for experimental validation.

4.3.1 Proportion of Causal Variants Identified vs. SNPs Selected

The primary evaluation metric is the following curve:

1. Rank all SNPs across all loci (that harbour at least one causal variant) by their posterior probabilities (or other prioritisation score).
2. For each threshold on the average number of SNPs selected per locus, compute the proportion of true causal variants captured.

Formally, for a given number of top SNPs T selected per locus:

$$\text{Proportion of causals identified} = \frac{\#\{\text{true causal variants among top-}T \text{ per locus}\}}{\#\{\text{total true causal variants}\}}. \quad (50)$$

Results are averaged over 500 simulation replicates. The key summary statistic is the average number of SNPs per locus required to capture 50% and 90% of all causal variants.

Comparison with other methods: This metric is similar to FINEMAP’s “proportion of causal SNPs included vs. number of top SNPs” but differs in that PAINTOR ranks SNPs *across all loci jointly* (using global posterior probabilities), not within each locus independently. This global ranking can shift probability mass between loci.

4.3.2 Credible Set Size

The size of the global ρ -level credible set (90%, 95%, 99%) is reported as the total number of SNPs required across all loci to capture ρ fraction of the total posterior mass. This is compared between PAINTOR with and without functional annotations.

4.3.3 Positive Predictive Value (Supplementary)

In supplementary analyses, PAINTOR also reports Positive Predictive Value (PPV):

$$\text{PPV} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad (51)$$

as a function of the number of SNPs selected. The authors note that the relative performance of all methods is maintained when using PPV instead of proportion of causals identified.

4.3.4 Enrichment Estimation Accuracy

PAINTOR evaluates the accuracy of its functional annotation enrichment estimates by comparing inferred $\log_2$ enrichment values against the true simulated enrichment, across a range from 8-fold depletion to 8-fold enrichment. Mean and standard deviation are reported over 500 simulations. This is compared with fgwas.

4.3.5 Cost-to-Benefit Optimisation

A unique feature of PAINTOR’s evaluation is a utility framework for determining the optimal number of SNPs to follow up:

$$U = B \cdot N_c - C \cdot N_t, \quad (52)$$

where B is the benefit per causal variant identified, C is the cost per SNP tested, N_c is the number of true causals found, and N_t is the total number of SNPs selected. The ratio $R = B/C$ is varied and the utility is plotted as a function of the posterior probability threshold.

4.3.6 Metrics *Not* Reported

- Per-signal credible set coverage, power, or purity (PAINTOR uses global credible sets, not per-signal sets).
- Power vs. FDR curves or AUPRC.
- PIP calibration plots.
- Credible set purity.

4.3.7 Differences from SuSiE’s, BEATRICE’s, and FINEMAP’s Evaluation

1. **Credible set definition.** PAINTOR uses *global* credible sets across all loci (SNPs ranked by PIP, cumulated to $\rho\%$ of total posterior mass). SuSiE uses *per-signal* credible sets. The two are not comparable.
2. **Primary metric.** PAINTOR reports “proportion of causal variants identified vs. average SNPs per locus.” SuSiE reports coverage, power, and size of per-signal credible sets plus power-vs-FDR curves.
3. **Functional annotation enrichment.** PAINTOR’s simulations are designed around annotation enrichment; the data-generating process uses the same logistic annotation model as the inference procedure. Other methods do not use functional annotations.
4. **NCP handling.** PAINTOR fixes NCPs to point estimates; FINEMAP/CAVIARBF integrate them out; SuSiE estimates the prior effect variance via empirical Bayes.
5. **Maximum causal variants.** PAINTOR restricts enumeration to ≤ 3 causals per locus; FINEMAP and SuSiE allow more ($K = 5$ or $L = 10$).
6. **Effect size distribution.** PAINTOR partitions total heritability equally among causal SNPs (deterministic effect sizes given h^2 and K); SuSiE draws from $\mathcal{N}(0, 0.6^2)$; BEATRICE draws from $\mathcal{N}(0, \mathbf{I}_d)$.
7. **Simulation scale.** 100 loci $\times$ 500 replicates; only loci with ≥ 1 causal variant are evaluated (~ 64 per simulation).

5 FiniMOM

5.1 Simulation Design

Karhunen et al. simulate fine-mapping data designed to mimic typical protein quantitative trait locus (pQTL) analyses. A key design feature is the use of *real* genotype data from two independent Finnish birth cohorts: one for generating summary statistics and one for providing an out-of-sample LD reference, enabling a direct comparison of in-sample vs. out-of-sample LD performance.

5.1.1 Genotype Data

Two Finnish population-based cohorts provide the genotype data:

- **NFBC1966** ($N = 5\,400$): used for phenotype simulation and computing summary statistics (and in-sample LD).
- **NFBC1986** ($N = 3\,743$): used for computing the out-of-sample LD matrix.

Five protein-coding genes of varying size are randomly selected from different chromosomes. For each gene, the fine-mapping region is defined as ± 100 Kb around the gene. Variants are filtered

for availability in both cohorts, yielding between $P = 387$ and $P = 2996$ variants per locus. Variants are clumped at $r^2 = 0.99$ (near-perfect duplicates removed), and an LD consistency check is applied when using the out-of-sample LD matrix.

Additional simulations use $N = 50\,000$ individuals from a synthetic HAPNEST genotype data set to assess performance at larger sample sizes.

5.1.2 Generative Model for the Phenotype

Phenotypes are simulated under a standard linear model. The simulation parameters are designed to represent plausible pQTL scenarios:

- Number of causal variants per locus: $d \in \{1, 2, 5\}$.
- Proportion of phenotypic variance explained (PVE) per locus: $h^2 \in \{0.015, 0.03\}$.

These PVE values are notably smaller than those used by SuSiE ($\phi \in \{0.05, 0.1, 0.2, 0.4\}$) and are intended to match the weaker signals typical of pQTL studies.

The paper does not provide explicit formulae for the effect size distribution or the phenotype-generating equation in the main text; these details are deferred to the Supplementary Methods. The summary statistics (coefficient estimates $\hat{\beta}_j$ and their standard errors $\text{SE}(\hat{\beta}_j)$) are computed from the NFBC1966 genotype data and provided as input to the fine-mapping methods.

5.1.3 Simulation Parameters

Each scenario (combination of d , h^2 , locus, and LD type) is repeated 100 times. The key parameters are:

$$d \in \{1, 2, 5\}, \tag{53}$$

$$h^2 \in \{0.015, 0.03\}, \tag{54}$$

$$P \in \{387, \dots, 2996\} \quad (\text{varies by locus}). \tag{55}$$

For FiniMOM, the hyperparameters are varied across a grid:

- $\tau \in \{0.00320, 0.00385, 0.00538, 0.0083\}$ (controls the spread of detectable effect sizes in the piMOM prior);
- $u \in \{1.05, 1.25, 1.5, 1.75, 2, 2.25, 2.5\}$ (controls the beta-binomial model dimension prior);
- $r = 1$ (tail parameter, giving Cauchy-like tails);
- $K = 10$ (maximum number of causal variants).

The default parameter values chosen for comparison with SuSiE are $\tau = 0.00385$, $u = 2$ (in-sample LD) or $u = 2.25$ (out-of-sample LD). SuSiE is run with $L = 10$ and σ_l^2 estimated from the data.

MCMC is run for 12 500 iterations with 2 500 burn-in, using the last 10 000 samples for posterior inference.

Note for benchmarking: The simulation PVE values ($h^2 \in \{0.015, 0.03\}$) are substantially lower than those used by SuSiE (ϕ up to 0.4) or BEATRICE (ω^2 up to 0.8). The use of two independent Finnish cohorts for in-sample vs. out-of-sample LD comparison is a unique and practically relevant design feature not present in the other papers.

5.2 Post-Processing of Results

FiniMOM’s post-processing produces per-variant PIPs and per-signal credible sets, following a procedure closely aligned with (but not identical to) SuSiE.

5.2.1 Posterior Inclusion Probabilities

PIPs are computed as the proportion of MCMC posterior samples in which variant j is included in the model:

$$\text{PIP}_j = P(\beta_j \neq 0 \mid \mathcal{D}) \approx \frac{1}{T} \sum_{t=1}^T \mathbf{1}(j \in \mathbf{m}^{(t)}), \quad (56)$$

where $\mathbf{m}^{(t)}$ is the model (set of included variants) at MCMC iteration t and $T = 10\,000$ is the number of post-burn-in samples.

5.2.2 Credible Set Construction

FiniMOM adopts the credible set definition of Lee et al. (2018) and Wang et al. (2020): a level- α credible set is a subset of variants that contains a true causal variant with probability $\geq \alpha$.

The number of credible sets is determined by the *posterior distribution of model dimension* $P(d = l \mid \mathcal{D})$ for $l = 1, \dots, K$, which is directly available from the MCMC output. This provides a posterior probability for each possible number of signals—a feature unique to FiniMOM among the methods reviewed here. SuSiE determines the number of credible sets via the purity filter, and the other methods use various heuristics.

Credible sets are constructed at coverage α_l , which represents the coverage of α conditioned on l signals. The details are described in the Supplementary Methods.

5.2.3 Purity Filtering

FiniMOM uses SuSiE’s purity filtering approach for discarding uninformative credible sets, following the convention of discarding sets with minimum absolute pairwise correlation below 0.5. However, the paper does not explicitly state the purity threshold used; results are compared directly with SuSiE’s credible sets.

5.2.4 Summary of the Pipeline

1. Optionally clump variants at $r^2 = 0.99$ and perform LD consistency checks (for out-of-sample LD).
2. Run MCMC (12 500 iterations, 2 500 burn-in) with the piMOM prior for effect sizes and beta-binomial prior for model dimension.
3. Compute PIPs as the proportion of posterior samples including each variant.
4. Determine the number of credible sets from the posterior distribution of model dimension.
5. Construct credible sets at the appropriate conditional coverage level.

5.3 Evaluation Metrics

FiniMOM uses the same three credible-set-based metrics as SuSiE, making it one of the most directly comparable methods to SuSiE in this benchmarking report.

5.3.1 95% Credible Set Coverage

The proportion of credible sets that contain at least one true causal variant:

$$\text{Coverage} = \frac{\#\{\text{CSs containing } \geq 1 \text{ true causal variant}\}}{\#\{\text{total CSs reported}\}}. \quad (57)$$

Reported as a mean over 100 simulation replicates, with 95% confidence intervals computed using the binomial distribution.

5.3.2 Power

The proportion of true causal variants captured by any reported credible set:

$$\text{Power} = \frac{\#\{\text{true causal variants in at least one CS}\}}{\#\{\text{total true causal variants}\}}. \quad (58)$$

Reported with 95% binomial confidence intervals.

5.3.3 Median Credible Set Size

The median number of variants per credible set, with 95% confidence intervals obtained via 1 000 bootstrap samples.

5.3.4 Additional Supplementary Metrics

The following are reported in supplementary figures:

- **PIP distributions and ranks** of true causal variants (Supplementary Figure S6).
- **Number of credible sets** detected (Supplementary Figure S7).
- **Computational time** as a function of the number of variants (Supplementary Figure S5).
- **Posterior distribution of model dimension** $P(d = l \mid \mathcal{D})$ —a unique output of FiniMOM providing uncertainty quantification on the number of causal signals.

5.3.5 Differences from Other Methods' Evaluation

1. **Metric alignment with SuSiE.** FiniMOM uses the same three core metrics as SuSiE (coverage, power, median size), making direct comparison straightforward. However, FiniMOM does *not* report power-vs-FDR curves or PIP calibration plots.
2. **Confidence intervals.** FiniMOM explicitly reports 95% CIs for all metrics (binomial for coverage/power, bootstrap for median size). SuSiE reports ± 2 standard errors for coverage only.
3. **In-sample vs. out-of-sample LD.** FiniMOM systematically compares performance under both LD settings, which is not done in the original SuSiE paper (though it is addressed in the SuSiE-RSS paper by Zou et al. 2022).
4. **Model dimension posterior.** FiniMOM uniquely provides the posterior distribution over the number of causal signals, which is not available from SuSiE, FINEMAP, BEAT-RICE, or PAINTOR.
5. **No purity metric.** FiniMOM does not report average r^2 or minimum absolute correlation for credible sets.
6. **Simulation scale.** 5 loci $\times$ 6 scenarios $\times$ 100 replicates = 3 000 simulations. This is smaller than SuSiE (6 000 data sets) but each simulation involves a full MCMC run.

7. **PVE range.** FiniMOM simulates at $h^2 \in \{0.015, 0.03\}$, which is much lower than SuSiE (ϕ up to 0.4) or BEATRICE (ω^2 up to 0.8), reflecting the pQTL application focus.
8. **No FDR curves, AUPRC, or calibration.** These are not provided.

6 PolyFun

6.1 Simulation Design

Weissbrod et al. simulate fine-mapping data at a genome-wide scale using real UK Biobank genotypes and a realistic polygenic generative model. PolyFun is not itself a fine-mapping method but a *framework* that estimates functionally informed prior causal probabilities and provides them as input to downstream fine-mapping methods (SuSiE or FINEMAP). The simulations therefore evaluate the combined pipeline (e.g., PolyFun + SuSiE) rather than PolyFun in isolation.

6.1.1 Genotype Data

Real genotype data from the UK Biobank ($N \approx 320\,000$ British individuals) are used. Approximately 18.2 million imputed SNPs with $\text{MAF} \geq 0.001$ and INFO score ≥ 0.6 are included (excluding three long-range LD regions). In-sample LD is computed directly from the target UK Biobank samples. Additional simulations with mismatched reference LD are also conducted.

6.1.2 Generative Model for the Phenotype

Traits are simulated with a genome-wide polygenic architecture:

- Total heritability: $h^2 = 0.25$.
- Genome-wide proportion of causal SNPs: 0.5% of all SNPs.
- Each target fine-mapping locus contains approximately 10 causal SNPs.

Causal SNP effect sizes are drawn from a distribution where per-SNP heritabilities depend on functional annotations, using the baseline-LF 2.2.UKB model (187 annotations including coding, conserved, regulatory, MAF, and LD-related annotations). This means causal variants are more likely to reside in functionally enriched regions, and their effect sizes are scaled according to the annotation-dependent per-SNP heritability estimates from S-LDSC.

The phenotype is then generated as a linear combination of all causal SNP effects plus Gaussian noise, with the total variance normalised to achieve the target heritability.

Key distinction: Unlike the purely sparse simulations of SuSiE, FINEMAP, or FiniMOM (where only a handful of SNPs per locus are causal and the rest have zero effect), PolyFun’s simulations are *genome-wide polygenic*. With 0.5% of ~ 18 million SNPs being causal, there are approximately 90 000 causal SNPs genome-wide, with ~ 10 per target locus. This realistic polygenic architecture means that non-causal SNPs in LD with genome-wide causal variants will have inflated statistics even outside the target locus, creating a more challenging (and realistic) fine-mapping scenario.

6.1.3 Functional Annotation Enrichment

Causal variant prior probabilities are set proportional to per-SNP heritability estimates from the baseline-LF model. This generates a simulation where causal variants are genuinely enriched in biologically relevant annotations (e.g., coding exons, conserved regions, enhancers). The enrichment structure is not synthetic (as in PAINTOR) but derived from the actual annotation model fitted to UK Biobank data.

6.1.4 PolyFun Prior Estimation Pipeline

PolyFun specifies prior causal probabilities through a five-step procedure:

1. Estimate functional enrichments for 187 baseline-LF annotations using an L_2 -regularised extension of S-LDSC, restricted to even (respectively, odd) chromosomes.
2. Estimate per-SNP heritabilities for SNPs on odd (respectively, even) chromosomes using the enrichment estimates from Step 1.
3. Partition all SNPs into 20 bins of similar estimated per-SNP heritability.
4. Re-estimate per-SNP heritabilities within each bin, excluding the target chromosome (to avoid overfitting).
5. Specify prior causal probabilities in proportion to the re-estimated per-SNP heritabilities.

These prior probabilities are then provided as input to SuSiE (with $L = 10$) or a modified version of FINEMAP (v1.4, extended to accept prior causal probabilities).

6.1.5 Simulation Parameters and Compared Methods

Ten fine-mapping methods are compared, including: SuSiE, PolyFun + SuSiE, FINEMAP, PolyFun + FINEMAP, fastPAINTOR, CAVIARBF (with 1 or 2 maximum causal variants), and others. The number of simulation replicates and the exact locus definitions are detailed in the Supplementary Note but are not fully specified in the main text accessible from the abstract.

6.2 Post-Processing of Results

PolyFun itself does not produce PIPs or credible sets. Instead, it provides functionally informed prior causal probabilities, which are passed to downstream methods (SuSiE or FINEMAP). The post-processing therefore inherits the pipeline of the downstream method:

- **PolyFun + SuSiE:** PIPs are computed using SuSiE’s standard formula (Equation 3.12 of the SuSiE paper), and credible sets are constructed via the cumulative- α_l procedure with purity filtering (minimum absolute correlation ≥ 0.5), exactly as in standard SuSiE.
- **PolyFun + FINEMAP:** A modified version of FINEMAP (v1.4) is introduced that accepts per-SNP prior causal probabilities. PIPs are computed by marginalisation over the stochastic search output Γ^* , as in standard FINEMAP.

The only “post-processing” unique to PolyFun is the upstream computation of prior causal probabilities (the five-step pipeline above). The downstream fine-mapping output is evaluated directly without additional transformations.

Note for benchmarking: Because PolyFun modifies the *input* to SuSiE/FINEMAP (the prior) rather than the output, the post-processing pipeline is identical to whichever downstream method is used. The improvement comes entirely from better-specified priors.

6.3 Evaluation Metrics

PolyFun’s evaluation is centred on two metrics: false discovery rate (FDR) calibration and power, both defined with respect to PIP thresholds. This framework is more aligned with SuSiE’s evaluation than with PAINTOR’s or FINEMAP’s.

6.3.1 FDR (Calibration)

For a given PIP threshold t (e.g., $t = 0.95$ or $t = 0.5$), the FDR is defined as the proportion of false positives among all SNPs with $\text{PIP} \geq t$:

$$\text{FDR}(t) = \frac{\#\{\text{SNPs with PIP} \geq t \text{ that are non-causal}\}}{\#\{\text{SNPs with PIP} \geq t\}}. \quad (59)$$

This is computed by aggregating results across all simulations. A conservative estimate of the FDR is obtained as $1 - t$ (i.e., at $t = 0.95$, the estimated FDR is ≤ 0.05). A method is “well-calibrated” if its empirical FDR does not significantly exceed this conservative estimate.

The authors present both a *conservative* (upper) and an *anti-conservative* (lower) FDR bound. Methods whose FDR significantly exceeds the conservative bound are flagged as having inflated false discovery rates.

6.3.2 Power

Power is defined as the proportion of true causal SNPs with PIP above a given threshold, aggregated across all simulations:

$$\text{Power}(t) = \frac{\#\{\text{true causal SNPs with PIP} \geq t\}}{\#\{\text{total true causal SNPs}\}}. \quad (60)$$

Power is reported at two thresholds: $\text{PIP} \geq 0.95$ and $\text{PIP} \geq 0.5$. This differs from SuSiE’s definition of power (proportion of causal variants captured by any credible set) but is more directly comparable to the PIP-based FDR–Power curves.

6.3.3 Computational Cost

Average runtime to fine-map a 3 Mb locus is reported on a log scale for all methods, both with and without functionally informed priors. Total runtime for fine-mapping different numbers of loci is also reported. PolyFun + SuSiE and PolyFun + FINEMAP runtimes are sub-linear because they include the fixed preprocessing cost of computing prior causal probabilities (~ 630 minutes).

6.3.4 Perturbation Analysis

A unique evaluation feature is the *perturbation analysis* of each of the five PolyFun steps. For each step, the authors introduce a specific perturbation (e.g., randomly shuffling annotation coefficients, varying the number of heritability bins) and assess the impact on FDR and power. This provides a detailed understanding of which components of the pipeline are critical.

6.3.5 Polygenic Localization (PolyLoc)

Beyond standard fine-mapping evaluation, PolyFun introduces a novel metric called *polygenic localization*. Using posterior mean per-SNP heritabilities from PolyFun + SuSiE, minimal sets of common SNPs that causally explain a given proportion (e.g., 50%) of common SNP heritability are constructed. These sets range from 28 SNPs (hair colour) to 3 400 (height) to ~ 2 million (number of children), providing a trait-level summary of genetic architecture complexity.

6.3.6 Real Data Application (49 UK Biobank Traits)

PolyFun + SuSiE is applied to 49 UK Biobank traits (average $N = 318\,000$), identifying 3 025 fine-mapped variant–trait pairs with $\text{PIP} > 0.95$ (a $> 32\%$ improvement over SuSiE alone). Functional enrichment of fine-mapped SNPs in annotations (coding, conserved, regulatory) is

reported as the ratio of the proportion of fine-mapped SNPs in an annotation to the proportion of all genome-wide SNPs in that annotation, stratified by PIP bins.

6.3.7 Differences from Other Methods’ Evaluation

1. **Framework vs. method.** PolyFun is a prior-specification framework, not a standalone fine-mapping method. It must be evaluated in combination with a downstream method (SuSiE or FINEMAP). This means its “improvements” reflect the value of better priors, not a different inference algorithm.
2. **Genome-wide polygenic simulations.** Unlike all other methods reviewed here, PolyFun simulates a realistic genome-wide polygenic architecture (0.5% causal SNPs, $\sim 90\,000$ genome-wide). This creates polygenic background noise absent from the sparse simulations of SuSiE, FINEMAP, BEATRICE, FiniMOM, and PAINTOR.
3. **FDR calibration as primary metric.** PolyFun emphasises FDR calibration at specific PIP thresholds, whereas SuSiE emphasises credible set coverage. Both assess calibration but in different ways.
4. **Power definition.** PolyFun defines power as the proportion of causal SNPs exceeding a PIP threshold; SuSiE defines power as the proportion of causal SNPs captured by any credible set. These are related but not identical.
5. **No credible set metrics.** The main paper does not report credible set coverage, size, or purity. Evaluation is entirely PIP-based.
6. **Functional annotation realism.** PolyFun uses 187 real biological annotations from the baseline-LF model; PAINTOR uses 3 synthetic annotations; other methods use no annotations.
7. **Scale.** PolyFun operates at genome-wide scale (~ 18 million SNPs, 49 traits, $N \approx 318\,000$), far larger than any other method’s evaluation.
8. **Perturbation analysis.** Unique to PolyFun; no other method systematically evaluates the robustness of each pipeline step.

7 SparsePro

7.1 Simulation Design

Zhang et al. evaluate SparsePro using both locus-level and genome-wide simulations based on real UK Biobank genotype data. The simulations incorporate functional annotations with controlled enrichment levels, enabling assessment of both annotation-free and annotation-informed fine-mapping.

7.1.1 Genotype Data

Real genotype data from $N = 353\,570$ unrelated UK Biobank White British ancestry individuals are used. For locus simulations, three 1 Mb regions are randomly selected. The in-sample LD matrix is computed from the UK Biobank genotypes.

7.1.2 Locus Simulations

For each of the three selected 1 Mb loci, 50 replicates are generated per combination of parameters:

- $K \in \{1, 2, 5, 10\}$: the number of causal variants per locus.

- $W \in \{0, 1, 2\}$: the enrichment intensity for functional annotations.

Causal variants are randomly assigned in each replicate. Five functional annotations are used: “conserved sequences,” “DNase I hypersensitive sites” (DHS), “non-synonymous,” and two histone marks (H3K27ac, H3K4me3). The enrichment weight vector $\mathbf{w}$ is set so that entries corresponding to the enriched annotations have value W (with $W = 0$ meaning no enrichment, i.e., a flat prior).

Phenotypes are simulated using the GCTA GWAS simulation pipeline with a continuous trait model and total heritability of $K \times 10^{-4}$ per locus (i.e., heritability scales linearly with the number of causal variants: 10^{-4} per causal variant). Association testing is then performed using GCTA to obtain GWAS summary statistics (z -scores or $\hat{\beta}$ and standard errors).

This yields a total of $3 \times 50 \times 4 \times 3 = 1\,800$ simulation replicates for locus simulations.

7.1.3 Genome-Wide Simulations

Genome-wide simulations are also conducted to compare SparsePro with PolyFun. Details follow a similar structure but at a genome-wide scale, with enrichment levels $W \in \{0, 1, 2\}$ and tests for annotation misspecification (using the “conserved sequences” annotation as a misspecified proxy when the true enrichment is in “non-synonymous” variants). Specific genome-wide simulation parameters (number of causal SNPs, total heritability) are described in the supplementary materials.

7.1.4 Compared Methods

SparsePro is evaluated in two configurations:

- **SparsePro**–: without functional annotations (annotation-free fine-mapping).
- **SparsePro**+: with functional annotations (annotation-informed fine-mapping).

Comparisons are made against: SuSiE ($L = 10$, σ_l^2 estimated from data), FINEMAP, PAINTOR– (without annotations), PAINTOR+ (with annotations), and PolyFun + SuSiE (genome-wide simulations only).

Note for benchmarking: The per-causal-variant heritability of 10^{-4} is very small—similar to FiniMOM’s pQTL-focused simulations and much smaller than SuSiE’s PVE settings (0.05–0.4). The number of causal variants (K up to 10) is higher than in most other benchmarks. The use of real UK Biobank genotypes with in-sample LD and GCTA for phenotype simulation provides a realistic evaluation framework.

7.2 Post-Processing of Results

SparsePro extends SuSiE’s IBSS algorithm with modified hyperparameter estimation and an alternative credible set construction strategy.

7.2.1 Posterior Inclusion Probabilities

SparsePro uses the same sum-of-single-effects decomposition as SuSiE. Each “effect group” k ($k = 1, \dots, K$) has a sparse indicator vector $\mathbf{s}_k$ drawn from a categorical distribution with prior probabilities $\boldsymbol{\pi}$. The PIP for variant j is computed identically to SuSiE:

$$\text{PIP}_j \approx 1 - \prod_{k=1}^K (1 - \alpha_{kj}), \quad (61)$$

where α_{kj} is the posterior probability that effect group k selects variant j .

7.2.2 Hyperparameter Estimation

A key difference from SuSiE is the use of *local heritability-based estimates* from HESS to set the hyperparameters for effect size precision (τ_β) and residual error precision (τ_y), rather than estimating them within the IBSS loop via the ELBO. This reduces the number of parameters estimated during fine-mapping and is reported to improve power and efficiency.

7.2.3 Credible Set Construction: Attainable Coverage

SparsePro introduces an *attainable coverage-based* approach for credible set construction, which differs from SuSiE’s purity-based filtering:

1. For each effect group k , compute the “attainable coverage”—the maximum coverage that can be achieved by a credible set for that effect group.
2. Only effect groups with attainable coverage $\geq \rho$ (e.g., 0.95) are summarised into ρ -level credible sets.
3. Credible sets are constructed by sorting variants by α_{kj} and accumulating until the cumulative probability reaches ρ .

This replaces SuSiE’s strategy of forming all L credible sets and then discarding those with purity < 0.5 . The authors show that applying the attainable coverage approach to SuSiE also improves its set-level summaries.

7.2.4 Functional Annotation Integration

When functional annotations are provided (SparsePro+), the prior inclusion probabilities are set via a softmax model:

$$\pi_g = \frac{\exp(\mathbf{A}_g^T \mathbf{w})}{\sum_{g'} \exp(\mathbf{A}_{g'}^T \mathbf{w})}, \quad (62)$$

where $\mathbf{A}_g$ is the annotation vector for variant g and $\mathbf{w}$ is the enrichment weight vector estimated jointly from the data. A G-test is introduced to screen annotations for relevance before incorporation (annotations with $p > 10^{-5}$ are excluded).

7.2.5 Summary of the Pipeline

1. Estimate hyperparameters (τ_β, τ_y) from local heritability estimates (HESS).
2. Optionally: estimate annotation enrichment weights $\mathbf{w}$ and apply G-test to select relevant annotations.
3. Run the paired mean field variational inference algorithm (equivalent to SuSiE’s IBSS).
4. Compute PIPs by marginalisation over effect groups.
5. Construct credible sets using the attainable coverage criterion.

7.3 Evaluation Metrics

SparsePro uses a comprehensive set of evaluation metrics that combines elements from both SuSiE-style and PAINTOR-style evaluations.

7.3.1 AUPRC (Area Under the Precision–Recall Curve)

The primary PIP-based metric is the AUPRC, computed by sweeping a threshold on the PIPs and calculating precision and recall against the true causal status:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad (63)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (64)$$

This is the same metric used by BEATRICE.

7.3.2 PIP Calibration

PIP calibration is assessed by grouping variants into five bins by their PIP values and plotting actual precision (proportion of true causal variants in each bin) against expected precision (mean PIP in the bin). A well-calibrated method should produce points close to the identity line. This provides a more granular calibration check than SuSiE’s supplementary calibration plots.

7.3.3 95% Credible Set Coverage

The proportion of credible sets containing at least one true causal variant:

$$\text{Coverage} = \frac{\#\{\text{CSs containing } \geq 1 \text{ true causal variant}\}}{\#\{\text{total CSs reported}\}}. \quad (65)$$

7.3.4 Credible Set Power

The proportion of true causal variants captured by any credible set:

$$\text{Power} = \frac{\#\{\text{true causal variants in at least one CS}\}}{\#\{\text{total true causal variants}\}}. \quad (66)$$

7.3.5 Credible Set Size

The *mean* number of variants per credible set (note: SuSiE reports *median* size; SparsePro reports *mean* size).

7.3.6 Computational Time

Runtime in seconds is reported per locus, broken down by number of causal variants. SparsePro is shown to scale favourably compared to FINEMAP, whose cost increases sharply with K .

7.3.7 Enrichment Weight Estimation

The accuracy of annotation enrichment weight estimation is evaluated by comparing estimated weights to true simulated values. The logarithmic relative ratio (logRR) between the largest and smallest prior inclusion probabilities is used to assess how well the method adapts to actual functional enrichment levels. This is compared against PolyFun’s annotation coefficient estimates from S-LDSC.

7.3.8 Annotation Misspecification Robustness

A unique evaluation: SparsePro is tested under annotation misspecification, where the wrong annotation (“conserved sequences”) is provided instead of the truly enriched one (“non-synonymous”). SparsePro is shown to be robust (attenuated but not inflated enrichment weights), while PolyFun generates strong priors from the misspecified annotation, leading to inflated false positives.

7.3.9 Differences from Other Methods' Evaluation

1. **Metric alignment.** SparsePro reports the same three credible set metrics as SuSiE (coverage, power, size) plus AUPRC and PIP calibration, making it one of the most comprehensive evaluations.
2. **Mean vs. median size.** SparsePro reports mean credible set size; SuSiE reports median. This can lead to different impressions when distributions are skewed.
3. **Attainable coverage vs. purity filtering.** SparsePro uses attainable coverage to determine which credible sets to report; SuSiE uses minimum pairwise correlation (purity ≥ 0.5). These are different selection criteria and may retain different sets.
4. **Annotation robustness testing.** SparsePro uniquely tests robustness to annotation misspecification, a practically important concern not addressed by PAINTOR or PolyFun's original evaluation.
5. **Heritability per locus.** SparsePro uses $K \times 10^{-4}$ heritability per locus, much smaller than SuSiE's $\phi \in \{0.05, 0.4\}$.
6. **No purity metric.** SparsePro does not report purity (minimum absolute correlation or average r^2) for credible sets.
7. **No FDR curves.** Power-vs-FDR curves are not provided; AUPRC serves as the aggregate PIP quality measure.

8 SBayesRC

8.1 Simulation Design

SBayesRC is primarily a *polygenic prediction* method that integrates GWAS summary statistics with functional genomic annotations. Unlike the other methods in this report, SBayesRC was not designed specifically for fine-mapping; rather, it is a genome-wide Bayesian multiple regression model (GBMM) that jointly fits all common SNPs and produces posterior inclusion probabilities (PIPs) as a by-product. The original method paper focuses on prediction accuracy, with fine-mapping evaluation being secondary. A more comprehensive fine-mapping evaluation using SBayesRC is provided in the follow-up preprint by Zeng et al. (2024).

8.1.1 Genotype and Phenotype Simulation

Simulations in the original SBayesRC paper use real UK Biobank genotype data with approximately 7 million common imputed SNPs and 96 functional annotations (from the baseline-LF model). Key simulation parameters:

- Sample size: N from UKB (exact size varies; LD reference samples range from $n = 500$ to $n = 20\,000$).
- Total heritability: $h^2 \in \{0.1, 0.5\}$.
- Polygenicity: approximately 1% of SNPs are causal ($\sim 70\,000$ causal variants genome-wide).
- Number of replicates: 10 independent simulations per setting.

Causal variants are drawn genome-wide with probabilities informed by functional annotations, and effect sizes are drawn from a mixture of zero and multiple normal distributions (the SBayesR/SBayesRC mixture prior). The phenotype is then generated as the sum of genetic effects plus Gaussian noise.

This is a *fully polygenic* generative model—far more causal variants ($\sim 70\,000$) than in any locus-based fine-mapping simulation (SuSiE: 1–5; FINEMAP: 5; BEATRICE: 1–12). The simulations are primarily designed to evaluate prediction accuracy and robustness to LD reference quality, not fine-mapping resolution.

8.1.2 LD Reference Robustness

A distinctive feature of SBayesRC’s simulation evaluation is the systematic comparison of LD reference panels of varying quality:

- **ukb20k**: 20 000 unrelated UKB European individuals (high quality).
- **uk10k**: 3 642 unrelated European individuals from the UK10K project.
- **1kg0.5k**: 494 unrelated European individuals from 1000 Genomes (low quality).
- **afr4k**: 4 000 African-ancestry UKB individuals (ancestry mismatch).

SBayesRC is shown to be robust to moderate LD mismatch but exhibits PIP inflation with very small reference samples ($n = 500$).

Note for benchmarking: The SBayesRC simulations are designed for the polygenic prediction use case. Fine-mapping metrics (PIPs, credible sets) are secondary outputs and are not the focus of the original paper’s evaluation. The follow-up paper by Zeng et al. (2024, medRxiv) provides the detailed fine-mapping benchmarking using SBayesRC.

8.2 Post-Processing of Results

SBayesRC produces PIPs as a by-product of its MCMC posterior inference over the genome-wide mixture model. The post-processing pipeline is fundamentally different from locus-based methods.

8.2.1 Model and MCMC Inference

SBayesRC uses a Bayesian mixture model where each SNP’s effect size is drawn from a mixture of a point mass at zero and C normal distributions with different variances:

$$\beta_j \sim \pi_0 \delta_0 + \sum_{c=1}^C \pi_c \mathcal{N}(0, \sigma_c^2), \quad (67)$$

where $\pi_0 + \sum_c \pi_c = 1$ and the mixture proportions π_c and component variances σ_c^2 can depend on functional annotations. Inference is performed via MCMC, with the posterior sampled genome-wide across all ~ 7 million SNPs simultaneously using a low-rank LD model based on eigen-decomposition of quasi-independent LD blocks.

8.2.2 Posterior Inclusion Probabilities

The PIP for each SNP is computed as the proportion of MCMC iterations in which the SNP is assigned to a non-null mixture component:

$$\text{PIP}_j = P(\beta_j \neq 0 \mid \text{data}) \approx \frac{1}{T} \sum_{t=1}^T \mathbf{1}(\beta_j^{(t)} \neq 0), \quad (68)$$

where T is the number of post-burn-in MCMC samples. Convergence is monitored via the potential scale reduction factor (PSRF); SNPs with high PIP but large PSRF should be interpreted with caution.

8.2.3 Credible Sets: Local Credible Sets (LCS)

The original SBayesRC paper does not describe credible set construction. The follow-up paper by Zeng et al. (2024) introduces *local credible sets* (LCS) for SBayesRC:

1. Identify candidate causal variants as SNPs with PIP above a threshold.
2. For each candidate, define a local region of SNPs in LD (using an LD r^2 threshold).
3. Construct a credible set by sorting SNPs in the local region by PIP and accumulating until the desired coverage is reached.

This LCS approach differs from SuSiE’s per-signal credible sets (which arise from the single-effect decomposition) and is closer to a post-hoc LD-clumping-based construction.

8.2.4 No Purity Filtering in Original Paper

The original SBayesRC paper does not apply purity filtering to credible sets (since it does not construct them). The follow-up paper uses the LD threshold in the LCS definition as an implicit form of purity control.

8.2.5 Summary of the Pipeline

1. Run SBayesRC MCMC genome-wide, jointly fitting ~ 7 million SNPs with 96 functional annotations.
2. Extract PIPs from MCMC posterior samples.
3. (Follow-up paper only) Construct local credible sets using LD-based grouping of high-PIP SNPs.
4. Output joint SNP effect estimates for polygenic score calculation.

8.3 Evaluation Metrics

The evaluation in the original SBayesRC paper is primarily focused on *polygenic prediction accuracy*, not fine-mapping metrics. However, several fine-mapping-relevant metrics are reported.

8.3.1 Prediction Accuracy (R^2)

The primary evaluation metric is the prediction R^2 (squared correlation between predicted and observed phenotype in a held-out sample), assessed via 10-fold cross-validation in UKB. This is compared across SBayesR, SBayesRC, LDpred2, LDpred-funct, MegaPRS, PolyPred-S, and PRS-CSx.

8.3.2 PIP Calibration

PIP calibration is assessed by binning SNPs by their PIP values and comparing the observed proportion of true causal variants in each bin to the expected proportion (mean PIP). Results are reported for simulations with $h^2 = 0.5$ and $h^2 = 0.1$. The authors show that SBayesRC PIPs are well-calibrated.

8.3.3 Effect Size Estimation Accuracy

Correlations between estimated and true effect sizes are reported for SNPs with PIP above different thresholds. This is a metric unique to SBayesRC among the methods reviewed here and reflects its joint-modelling approach.

8.3.4 Genetic Architecture Estimation

SBayesRC estimates SNP-based heritability and polygenicity (proportion of causal variants) per functional annotation. The accuracy of these estimates is compared to LDpred2 in simulations.

8.3.5 Metrics from the Follow-Up Paper (Zeng et al. 2024)

The follow-up genome-wide fine-mapping paper provides a much more comprehensive fine-mapping evaluation of SBayesRC, including:

- PIP calibration across different genetic architectures.
- Fine-mapping power: proportion of true causal SNPs with PIP above a threshold.
- Mapping precision: proportion of high-PIP SNPs that are truly causal.
- Credible set size (local credible sets).
- Replication rate in independent GWAS samples.
- Comparison with SuSiE, FINEMAP, and other region-specific methods.

These metrics are *not* in the original SBayesRC paper cited here but are important context for benchmarking.

8.3.6 Metrics *Not* Reported in the Original Paper

- Credible set coverage, power, or size (introduced only in the follow-up paper).
- Power-vs-FDR curves or AUPRC.
- Credible set purity.
- Comparison with other fine-mapping methods (the original paper compares only with polygenic prediction methods).

8.3.7 Differences from Other Methods' Evaluation

1. **Primary purpose.** SBayesRC is a polygenic prediction method; fine-mapping is a secondary output. All other methods reviewed are designed primarily for fine-mapping.
2. **Genome-wide joint modelling.** SBayesRC fits all ~ 7 million SNPs simultaneously via MCMC, rather than analysing individual loci. This means PIPs account for genome-wide LD and polygenic background, unlike locus-based methods (SuSiE, FINEMAP, PAINTOR, etc.) which analyse each locus independently.
3. **Mixture model prior.** SBayesRC uses a multi-component mixture prior (point mass + multiple normals), which explicitly models different effect size classes. This is richer than SuSiE's single normal prior per component or FINEMAP's single normal prior.
4. **Scale.** ~ 7 million SNPs, 96 annotations, 50 traits—far larger than any locus-based fine-mapping evaluation.
5. **Prediction focus.** The primary metrics (R^2 , prediction bias) are irrelevant to fine-mapping benchmarking. Fine-mapping metrics are secondary.
6. **No locus-level credible sets in original paper.** Credible sets and fine-mapping comparisons appear only in the follow-up paper (Zeng et al. 2024).

7. **LD robustness.** SBayesRC uniquely evaluates robustness to LD reference quality (sample size and ancestry mismatch), which is practically important but rarely tested systematically by other methods (except FiniMOM).
8. **Effect size accuracy.** SBayesRC reports correlation between estimated and true effect sizes at high-PIP SNPs—a metric not provided by any other method reviewed here.
9. **MCMC convergence.** SBayesRC reports PSRF for convergence monitoring; other MCMC-based methods (FiniMOM) do not explicitly report this.

Important caveat for benchmarking: If including SBayesRC in a fine-mapping benchmarking report, it is essential to note that (a) it was not designed for this purpose, (b) the original paper’s evaluation focuses on prediction, and (c) the comprehensive fine-mapping evaluation is in the follow-up Zeng et al. (2024) medRxiv preprint. Results from the original paper should be cited with this context.

9 CARMA

9.1 Simulation Design

Yang et al. simulate fine-mapping data using genotypes derived from 1000 Genomes haplotypes, with simulations conducted across 94 loci corresponding to real breast cancer GWAS risk regions. CARMA’s simulations are designed to evaluate performance under both standard (in-sample LD) and meta-analysis (external LD with discrepancies) settings.

9.1.1 Genotype Simulation

Genotypes are simulated using the R package `sim1000G`, based on 1000 Genomes Project Phase 3 European population data. The 94 loci are chosen to correspond to risk regions from a breast cancer GWAS, providing realistic size and LD structure. Rare variants ($\text{MAF} < 0.01$) are filtered out, leaving approximately 1 500–4 000 variants per locus. Genotype data are simulated for $n = 10\,000$ individuals.

9.1.2 Causal SNP Selection with Functional Annotations

Prior causal probabilities are generated using 200 DeepSEA chromatin features. Of these, 100 are randomly selected as being related to causal status, with their coefficients sampled from $\mathcal{N}(0, 0.1^2)$ (weak individual effects). The prior causal probability for each SNP is set proportional to a linear combination of annotations:

$$\Pr(\gamma_i = 1) \propto \mathbf{w}_i^T \boldsymbol{\theta}, \quad (69)$$

where $\mathbf{w}_i$ is the 201-dimensional annotation vector (including intercept) and $\boldsymbol{\theta}$ is the coefficient vector. Causal SNPs are selected from among those with the highest prior probabilities, subject to the constraint that the absolute pairwise Pearson correlation between any two causal variants is at most 0.1.

9.1.3 Generative Model for the Phenotype

For each causal SNP $i \in T$ (the true causal set), effect sizes are drawn independently from:

$$\beta_i \sim \mathcal{N}(0, 0.5^2), \quad \text{for } i \in T, \quad (70)$$

and $\beta_i = 0$ for all $i \notin T$. The phenotypic variance σ_y^2 is computed to achieve a per-locus PVE of $\phi = 0.0075$:

$$\phi = \frac{\text{Var}(\mathbf{X}\boldsymbol{\beta})}{\sigma_y^2 + \text{Var}(\mathbf{X}\boldsymbol{\beta})} = 0.0075. \quad (71)$$

The phenotype is then generated as $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, with $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_y^2 \mathbf{I}_n)$.

The number of causal variants per locus is varied: $|T| \in \{1, 2, 3\}$, with results for $|T| = 2$ shown in the main manuscript and $|T| \in \{1, 3\}$ in the supplement.

9.1.4 Meta-Analysis Simulation (External LD)

A unique feature of CARMA’s evaluation is the simulation of meta-analysis settings with LD–summary statistic discrepancies:

- The $n = 10\,000$ simulated individuals are split into three non-overlapping groups ($n_1 = 1\,000$, $n_2 = 1\,500$, $n_3 = 7\,500$).
- Summary statistics are computed per group and meta-analysed using inverse-variance weighting (METAL).
- **Consistent meta-analysis:** all SNPs use all three groups (LD from external UKBB panel is the only source of discrepancy).
- **Inconsistent meta-analysis:** 75% of SNPs use all three groups; 15% use only the two smaller groups; 10% use only the smallest group. This creates sample-size heterogeneity across SNPs, mimicking real meta-analysis data.

LD is estimated from UK Biobank individuals of European descent (external to the simulated data).

9.1.5 Compared Methods and Settings

- **CARMA:** Shotgun stochastic search with spike-and-slab prior; $\eta = 1$ (truncated Poisson prior on model size); $\tau = 0.05^2$ (prior precision); maximum 10 causal variants per locus.
- **SuSiE:** $L = 10$; PolyFun prior causal probabilities used when annotations are included.
- **fastPAINTOR:** Maximum 2 causal variants per locus; top 10 annotations selected.

CARMA and fastPAINTOR are run at the chromosome level (aggregating loci); SuSiE is run per locus.

Note for benchmarking: The per-locus PVE of 0.0075 is very small—comparable to Fin-iMOM’s pQTL simulations but much smaller than SuSiE’s (0.05–0.4). The constraint that causal variants must have pairwise $|r| \leq 0.1$ ensures well-separated signals, which may simplify the fine-mapping problem relative to scenarios where causal variants are in moderate LD.

9.2 Post-Processing of Results

CARMA produces PIPs, credible sets, and a novel “credible model” output via its shotgun stochastic search algorithm.

9.2.1 Posterior Inclusion Probabilities

CARMA uses the shotgun stochastic search (SSS) algorithm to explore the posterior model space. The SSS evaluates the neighbourhood of the current configuration at each iteration (adding, deleting, or swapping a causal SNP), recording all visited models and their unnormalised posterior probabilities. The PIP for each SNP is computed by marginalisation over all recorded models:

$$\text{PIP}_j = \sum_{\boldsymbol{\gamma} \in \Gamma^*} \mathbf{1}(\gamma_j = 1) p(\boldsymbol{\gamma} \mid \mathbf{Z}, \boldsymbol{\Sigma}). \quad (72)$$

A key advantage of the SSS over standard MCMC is that it explores the *neighbourhood* of each visited model, which provides more consistent PIPs for perfectly correlated SNPs. CARMA reports a standard deviation of 0.06 for PIPs within groups of perfectly correlated SNPs, compared to 0.493 for fastPAINTOR.

9.2.2 Credible Set Construction

CARMA constructs credible sets following the SuSiE definition: the smallest subset of correlated variants (with pairwise correlation exceeding a threshold r) that has probability ρ or greater of containing at least one causal variant. Results are reported for both $\rho = 0.95$ and $\rho = 0.99$.

For SuSiE, credible sets from SuSiE’s own output are used directly. For CARMA and fastPAINTOR, credible sets are constructed post hoc using the same procedure as SuSiE (sorting by PIP, accumulating to ρ).

9.2.3 Credible Models (Novel Output)

CARMA introduces the concept of a *credible model*: the set of candidate models whose posterior odds relative to the leading model (highest posterior probability) are below a threshold (e.g., 3.2 or 10). The variants included in the credible model provide a complementary summary to credible sets. In simulations, credible models achieve higher power with smaller sets of selected variants than credible sets.

9.2.4 Outlier Detection for Meta-Analyses

CARMA includes a Bayesian hypothesis testing procedure for detecting and removing outlier SNPs—those with discrepancies between their z -scores and the external LD structure. Outliers are identified and can be removed before fine-mapping (by CARMA or other methods such as SuSiE). This procedure is integrated into the CARMA algorithm, unlike SuSiE’s separate diagnostic step.

9.2.5 Summary of the Pipeline

1. Optionally: run outlier detection to identify and remove SNPs with LD– z -score inconsistencies.
2. Run shotgun stochastic search with spike-and-slab (or Cauchy) prior and truncated Poisson model-size prior.
3. Optionally: jointly estimate annotation enrichment weights via EM.
4. Compute PIPs by marginalisation over recorded models.
5. Construct credible sets (sorting by PIP, accumulating to ρ).
6. Report credible models (top models within a posterior odds threshold of the leading model).

9.3 Evaluation Metrics

CARMA uses a comprehensive set of metrics covering both PIP-based and credible-set-based evaluation, with particular emphasis on meta-analysis robustness.

9.3.1 Credible Set Metrics

Four metrics are reported for credible sets (at both $\rho = 0.95$ and $\rho = 0.99$):

Power. The overall proportion of simulated causal variants included in any credible set:

$$\text{Power} = \frac{\#\{\text{true causal variants in at least one CS}\}}{\#\{\text{total true causal variants}\}}. \quad (73)$$

Coverage. The proportion of credible sets that contain a causal variant:

$$\text{Coverage} = \frac{\#\{\text{CSs containing } \geq 1 \text{ true causal variant}\}}{\#\{\text{total CSs reported}\}}. \quad (74)$$

Size. The number of variants per credible set.

Purity. The mean squared correlation of variants within a credible set. This is the same metric used in SuSiE’s comparison with DAP-G (average r^2), enabling direct comparison.

9.3.2 PIP-Based Power and FDR

Power and FDR are computed using “positive predictions,” i.e., SNPs with PIP above a given threshold:

$$\text{Power}(t) = \frac{\#\{\text{true causal SNPs with PIP} \geq t\}}{\#\{\text{total true causal SNPs}\}}, \quad (75)$$

$$\text{FDR}(t) = \frac{\#\{\text{non-causal SNPs with PIP} \geq t\}}{\#\{\text{SNPs with PIP} \geq t\}}. \quad (76)$$

These are evaluated across multiple PIP thresholds, producing power-vs-FDR curves.

9.3.3 Credible Model Metrics

For credible models, the same power and size metrics are reported but applied to the set of variants included in the credible model (rather than credible sets).

9.3.4 PIP Consistency for Perfectly Correlated SNPs

A unique metric: the standard deviation of (standardised) PIPs within groups of perfectly correlated SNPs. This evaluates whether the search algorithm distributes posterior mass evenly among indistinguishable variants. CARMA (SD = 0.06) substantially outperforms fastPAINTOR (SD = 0.493); SuSiE achieves identical PIPs by construction.

9.3.5 Singleton Credible Set Analysis (Meta-Analysis)

In the meta-analysis simulations, the number of credible sets containing exactly one SNP and their coverage are reported separately. This highlights the problem of false singleton credible sets that arise from LD- z -score discrepancies.

9.3.6 Functional Enrichment of Fine-Mapped SNPs

In the Alzheimer’s disease application, functional enrichment is computed as the proportion of SNPs in credible models lying in a functional category divided by the proportion of genome-wide SNPs in that category—the same definition used by PolyFun.

9.3.7 Differences from Other Methods' Evaluation

1. **Meta-analysis focus.** CARMA is the only method that systematically evaluates fine-mapping under LD-summary statistic discrepancies (consistent vs. inconsistent meta-analysis). This is a practically critical scenario not tested by SuSiE, FINEMAP, BEATRICE, SparsePro, or FiniMOM.
2. **Credible model output.** CARMA uniquely reports credible models (top configurations by posterior probability) as a complement to credible sets. No other method provides this.
3. **Purity reported.** CARMA reports mean r^2 purity for credible sets, matching SuSiE's DAP-G comparison metric.
4. **PIP consistency metric.** The SD of PIPs among perfectly correlated SNPs is a unique and informative metric.
5. **Two coverage levels.** Results are shown for both $\rho = 0.95$ and $\rho = 0.99$; most other methods report only 0.95.
6. **Annotation integration.** CARMA jointly models annotations and summary statistics via EM (like PAINTOR), rather than using a two-stage approach (like PolyFun + SuSiE).
7. **Effect size prior.** CARMA offers both spike-and-slab ($\mathcal{N}(0, \tau^{-1})$) and heavy-tailed Cauchy priors; other methods use a single prior family.
8. **Computational cost.** CARMA's runtime is substantially higher than SuSiE and BEATRICE (noted in the BEATRICE paper as being the slowest method due to its MCMC/shotgun search).
9. **Small per-locus PVE.** $\phi = 0.0075$ per locus, similar to FiniMOM and SparsePro, much smaller than SuSiE.
10. **Causal variant LD constraint.** Causal variants are constrained to have $|r| \leq 0.1$, ensuring well-separated signals. This may be less challenging than scenarios with causal variants in moderate/high LD.

10 Funmap

10.1 Simulation Design

Li et al. evaluate Funmap using real UK Biobank genotype data across 10 breast cancer risk regions, with a particular focus on the challenge of integrating *high-dimensional* functional annotations without inflating the false discovery rate. The simulations span a wide range of annotation dimensionalities from low ($m = 20$) to high ($m = 100$).

10.1.1 Genotype Data

Real genotype data from UK Biobank White British individuals are used, with sample sizes $n \in \{20\,000, 50\,000\}$. Ten risk regions identified in a breast cancer GWAS are selected, each comprising approximately $p = 700\text{--}4\,200$ variants.

10.1.2 Functional Annotation Simulation

Unlike CARMA and SparsePro (which use real annotations), Funmap uses *synthetic* continuous annotations. For each SNP j and annotation k , the annotation value A_{jk} is drawn from a standard normal distribution:

$$A_{jk} \sim \mathcal{N}(0, 1), \quad j = 1, \dots, p, \quad k = 1, \dots, m. \quad (77)$$

The number of annotations is varied: $m \in \{20, 50, 100\}$, covering low to high-dimensional settings. Annotation effects on causal probability are drawn as:

$$w_k \sim \mathcal{N}(0, 0.01), \quad k = 1, \dots, m, \quad (78)$$

so each individual annotation has a weak effect on causal probability. The prior causal probabilities are then computed via the softmax function:

$$\pi_j = \frac{\exp(\mathbf{A}_j^T \mathbf{w})}{\sum_{j'=1}^p \exp(\mathbf{A}_{j'}^T \mathbf{w})}. \quad (79)$$

10.1.3 Causal Variant Selection

$L_0 \in \{2, 3\}$ causal SNPs per locus are selected as the variants with the highest values of π_j , subject to the constraint that pairwise correlation between any two causal variants is less than 0.1 (following CARMA’s convention).

10.1.4 Generative Model for the Phenotype

Given the causal SNP set, the effect sizes are drawn from:

$$\mathbf{b}_0 \sim \mathcal{N}\left(\mathbf{0}, \frac{0.0075}{L_0} \cdot \mathbf{I}_{L_0}\right), \quad (80)$$

where 0.0075 is the per-locus heritability contributed by the target region. The phenotype is generated as:

$$\mathbf{y} = \mathbf{X}_0 \mathbf{b}_0 + \mathbf{e}, \quad \mathbf{e} \sim \mathcal{N}\left(\mathbf{0}, \frac{1 - 0.0075}{0.0075} \cdot \text{Var}(\mathbf{X}_0 \mathbf{b}_0) \cdot \mathbf{I}_n\right), \quad (81)$$

where $\mathbf{X}_0$ is the standardised genotype matrix of causal SNPs. Univariate linear regressions are then performed to obtain per-SNP z -scores.

For each setting (combination of n , m , L_0 , and locus), 50 replicates are generated across the 10 regions, yielding 500 replicates per configuration.

10.1.5 Compared Methods

- **Without annotations:** SuSiE ($L = 10$), CARMA ($\eta = 10$), PAINTOR (fastPAINTOR with MCMC flag).
- **With annotations:** Funmap, CARMA+anno, PAINTOR+anno.

Note for benchmarking: The use of synthetic continuous annotations (drawn from $\mathcal{N}(0, 1)$) is different from SparsePro (real binary annotations), CARMA (real DeepSEA features), and PAINTOR (synthetic binary annotations). The per-locus heritability of 0.0075 is small and comparable to CARMA and SparsePro. The constraint that causal variants have pairwise $|r| \leq 0.1$ follows CARMA’s convention.

10.2 Post-Processing of Results

Funmap extends SuSiE’s sum-of-single-effects framework by incorporating functional annotations through a random effects model with a three-stage variational inference algorithm.

10.2.1 Posterior Inclusion Probabilities

Funmap uses the same PIP formula as SuSiE, derived from the component-specific posterior probabilities $q(\gamma_{lj} = 1)$:

$$\text{PIP}_j = 1 - \prod_{l=1}^L (1 - q(\gamma_{lj} = 1)). \quad (82)$$

10.2.2 Annotation Integration via Random Effects

The key innovation is the use of a *component-specific random effects* model for annotation weights:

$$\pi_{lj} = \frac{\exp(\mathbf{A}_j^T \mathbf{w}_l)}{\sum_{j'} \exp(\mathbf{A}_{j'}^T \mathbf{w}_l)}, \quad \mathbf{w}_l \sim \mathcal{N}(\mathbf{0}, \sigma_{w_l}^2 \mathbf{I}_m). \quad (83)$$

This differs from SparsePro and PAINTOR (which use fixed annotation effects shared across components) and PolyFun (which estimates priors in a separate upstream step). The random effects formulation provides automatic shrinkage of redundant annotation weights, which is the mechanism for controlling FDR under high-dimensional annotations.

10.2.3 Three-Stage Fitting with Warm-Starts

Funmap uses a three-stage variational inference procedure:

1. **Stage 1:** Fit SuSiE without annotations to initialise effect size parameters and posterior $q(\tilde{\mathbf{b}}, \tilde{\gamma})$.
2. **Stage 2:** Fix $\tilde{\gamma}$ at its posterior mean from Stage 1 and estimate annotation-related parameters $\sigma_{w_l}^2$ and $q(\tilde{\mathbf{w}})$.
3. **Stage 3:** Run the full Funmap model using warm-started parameters from Stages 1–2.

This warm-start procedure prevents excessive components from overfitting high-dimensional annotations, which the authors identify as a key reason why CARMA+anno produces inflated FDR.

10.2.4 FDR Control

Funmap computes a local FDR for each SNP as $\text{fdr}(j) = 1 - \text{PIP}(j)$, and a global FDR by sorting local FDRs and averaging:

$$\text{FDR}(j) = \frac{\sum_{i=1}^j \text{fdr}_{(i)}}{j}, \quad (84)$$

where $\text{fdr}_{(i)}$ is the i -th smallest local FDR. A SNP is declared causal if $\text{FDR}(j) < \eta$ for a given threshold η .

10.2.5 Credible Set Construction

Funmap inherits SuSiE’s credible set construction: for each component l , variants are sorted by $q(\gamma_{lj} = 1)$ and accumulated until the coverage threshold ρ is reached. The paper reports credible set sizes in the real-data application but does not provide systematic credible set coverage/power/size metrics in the simulation evaluation (these are shown primarily via box plots of credible set size across traits).

10.2.6 Annotation Importance Score

A unique output is the annotation importance score, defined as:

$$\text{FI}_k = \max_l \mu_{w_{lk}}^2, \quad (85)$$

where $\mu_{w_{lk}}$ is the posterior mean of the k -th annotation weight in the l -th component. This captures the strongest evidence of annotation relevance across all causal signals.

10.2.7 Summary of the Pipeline

1. Stage 1: Fit SuSiE without annotations (warm-start).
2. Stage 2: Estimate annotation weight posteriors with fixed causal signal estimates.
3. Stage 3: Run full Funmap model with all parameters warm-started.
4. Compute PIPs, local/global FDR, and credible sets.
5. Compute annotation importance scores.

10.3 Evaluation Metrics

Funmap’s evaluation focuses heavily on FDR calibration under high-dimensional annotations—the central claim of the paper—alongside power and credible set comparisons.

10.3.1 FDR Calibration (Primary Metric)

The primary evaluation metric is the empirical FDR (eFDR) plotted against the expected FDR level:

$$\text{eFDR} = 1 - \frac{\#\{\text{true causal SNPs among putative causal SNPs}\}}{\#\{\text{putative causal SNPs}\}}, \quad (86)$$

where putative causal SNPs are those with global FDR below a given threshold η . A well-calibrated method should have eFDR close to the expected FDR level across all thresholds. This is plotted as eFDR vs. expected FDR curves for each method and annotation dimensionality ($m = 20, 50, 100$).

The key finding: Funmap is the *only* method that maintains calibrated FDR across all annotation dimensionalities. PAINTOR+anno shows severely inflated FDR that worsens with increasing m . CARMA+anno also shows inflation for $m \geq 50$.

10.3.2 Power

Power is defined as the proportion of true causal SNPs identified among the putative causal SNPs at a given FDR threshold. It is reported alongside FDR calibration to demonstrate that Funmap achieves comparable or better power while maintaining FDR control.

10.3.3 PIP Comparison (Scatter Plots)

PIPs from Funmap are plotted against those from SuSiE, CARMA+anno, and PAINTOR+anno in scatter plots, with true causal variants highlighted. This allows visual comparison of how methods redistribute posterior mass when annotations are included.

10.3.4 Credible Set Size

In the real-data application (4 lipid traits with 187 baseline-LF annotations), credible set sizes are compared via box plots across Funmap, CARMA+anno, and PAINTOR+anno.

10.3.5 Replication Rate

A unique real-data evaluation metric: variants identified by each method from the discovery GWAS (Global Lipids Genetics Consortium, GLGC) are tested for replication in an independent cohort. Two replication criteria are used:

1. The fraction of newly identified SNPs (not genome-wide significant in discovery) that reach $p < 5 \times 10^{-8}$ in the replication cohort.
2. The fraction of newly identified SNPs included in 95% credible sets generated from the replication GWAS with SuSiE.

Funmap achieves 15.5%–26.2% improvement in replication rate over the runner-up method.

10.3.6 Annotation Importance Scores

Feature importance scores (Equation 11) are reported as box plots across genomic regions and traits, providing insights into which annotations contribute most to fine-mapping.

10.3.7 Computational Time

CPU timings are compared across methods and annotation dimensionalities, showing that Funmap’s variational algorithm is substantially faster than CARMA’s MCMC-based approach.

10.3.8 Differences from Other Methods’ Evaluation

1. **FDR calibration focus.** Funmap’s primary claim is FDR control under high-dimensional annotations—a problem not systematically tested by any other method. CARMA and SparsePro test annotation robustness but not FDR calibration across varying m .
2. **Synthetic continuous annotations.** Unlike CARMA (real DeepSEA features), SparsePro (real binary annotations), or PAINTOR (synthetic binary), Funmap uses synthetic continuous annotations drawn from $\mathcal{N}(0, 1)$.
3. **Random effects vs. fixed effects.** Funmap’s component-specific random effects model for annotation weights is unique. CARMA uses penalised logistic regression; SparsePro uses a softmax with shared weights; PAINTOR uses a logistic model with fixed effects.
4. **Replication rate.** Funmap is the only method (alongside FiniMOM) that validates fine-mapping results using an independent replication cohort, providing external validation beyond simulation.
5. **No systematic credible set metrics in simulations.** Unlike SuSiE, SparsePro, FiniMOM, and CARMA, Funmap does not report systematic credible set coverage/power/size across simulation settings. Credible set size is only shown in real-data box plots.
6. **No PIP calibration plots.** Unlike SparsePro and PolyFun, Funmap does not provide PIP calibration curves (binned PIP vs. observed proportion of causals).
7. **Annotation importance.** The feature importance score is a unique output enabling interpretation of which annotations drive fine-mapping improvements.
8. **Per-locus heritability.** 0.0075, consistent with CARMA and SparsePro.

11 SuSiE-inf / FINEMAP-inf

11.1 Simulation Design

Cui et al. conduct large-scale simulations designed to diagnose which sources of model misspecification cause PIP miscalibration in existing fine-mapping methods (SuSiE, FINEMAP), and to evaluate the proposed SuSiE-inf and FINEMAP-inf methods. The simulations use real UK Biobank genotypes and incorporate multiple, layered sources of misspecification.

11.1.1 Genotype Data

Real genotype data from $N = 149\,630$ unrelated individuals of white British ancestry in the UK Biobank are used. The genotype matrix includes imputed variants at genome-wide significant loci identified from BOLT-LMM association mapping. The LD matrix is computed in-sample.

11.1.2 Generative Model for the Phenotype

The phenotype is simulated with both sparse (large-effect) and non-sparse (infinitesimal) causal components:

$$\mathbf{y} = \mathbf{X}(\boldsymbol{\beta} + \boldsymbol{\alpha}) + \boldsymbol{\epsilon}, \quad (87)$$

where:

- $\boldsymbol{\beta}$ is a sparse vector of large-effect causal variants;
- $\boldsymbol{\alpha}$ is a dense vector of infinitesimal effects, with $\alpha_j \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \tau^2)$;
- $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ is residual noise.

The total SNP heritability is fixed at $h^2 = 0.5$. The key simulation parameter is the proportion of h^2 explained by the non-sparse component (the “non-sparsity fraction”).

11.1.3 Effect Size Distribution for Sparse Component

Rather than drawing sparse effect sizes from a simple normal distribution, the authors use effect size distributions estimated from real SuSiE fine-mapping of UK Biobank height. This means the sparse effect sizes reflect realistic, empirically estimated genetic architecture rather than a parametric assumption.

11.1.4 Simulation Scenarios and Misspecification Sources

The paper systematically varies multiple sources of model misspecification (Table 1 of the paper). Key scenarios include:

Ideal simulation. No misspecification: uniform causal prior, no imputation noise, no stratification, no non-sparse effects. This serves as the calibration baseline.

Baseline misspecification. Imputation noise, realistic (non-uniform) effect size prior from SuSiE Height posterior, and 19 out of 20 principal components (PCs) corrected in GWAS, but no non-sparse effects.

Non-sparsity simulations (key scenarios).

- **Moderate non-sparsity:** Non-sparse effects explain 58% of h^2 , with 1% coverage (i.e., 1% of variants have non-zero infinitesimal effects). Missing causal variants included.
- **High non-sparsity:** 75% of h^2 from non-sparse effects, 1% coverage.
- **Very high non-sparsity:** 83% of h^2 from non-sparse effects, 1% coverage.

- **No large effects:** 100% of h^2 from non-sparse effects (no sparse causals).

Additional simulations vary the coverage of non-sparse effects: 0.5%, 1%, and 5% of all variants.

Stratification simulations. Population stratification is simulated by adding effects of 20 PCs to the phenotype, with varying multipliers and varying numbers of PCs corrected in the GWAS. Two association mapping methods are compared:

- BOLT-LMM: a linear mixed model that accounts for relatedness and population structure.
- OLS: ordinary least squares with no correction for population structure.

Other misspecification sources. Imputation noise and QC-induced missing causal variants are included in all non-ideal simulations.

11.1.5 Association Mapping and Fine-Mapping Pipeline

GWAS is performed using BOLT-LMM (or OLS in stratification simulations). Fine-mapping is then performed at genome-wide significant loci using SuSiE, FINEMAP, SuSiE-inf, and FINEMAP-inf. This mirrors the standard pipeline used in real UK Biobank analyses.

11.1.6 Key Design Features

- Simulations are conducted at a realistic scale ($N \approx 150K$, genome-wide), unlike the small-sample locus-level simulations of SuSiE, BEATRICE, FiniMOM, SparsePro, CARMA, and Funmap.
- The non-sparse component is the central innovation: it tests whether the *purely sparse assumption* underlying all standard fine-mapping methods is a valid model of real genetic architecture.
- The use of empirically estimated effect size distributions (from real SuSiE posteriors) rather than parametric priors avoids confounding model misspecification in the effect size prior with the non-sparsity effect.

11.2 Post-Processing of Results

SuSiE-inf and FINEMAP-inf extend the standard SuSiE and FINEMAP pipelines by adding an infinitesimal variance component τ^2 estimated within each fine-mapped locus.

11.2.1 Model

The model is:

$$\mathbf{y} = \mathbf{X}(\boldsymbol{\beta} + \boldsymbol{\alpha}) + \boldsymbol{\epsilon}, \quad \alpha_j \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \tau^2), \quad \epsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2). \quad (88)$$

The goal of fine-mapping is to infer the non-zero coordinates of the sparse component $\boldsymbol{\beta}$, marginalising over both the infinitesimal effects $\boldsymbol{\alpha}$ and the residual noise $\boldsymbol{\epsilon}$.

11.2.2 FINEMAP-inf

For FINEMAP-inf, the sparse effects follow a point-normal prior:

$$\beta_j \stackrel{\text{iid}}{\sim} \pi_0 \mathcal{N}(0, s^2) + (1 - \pi_0) \delta_0. \quad (89)$$

Posterior inference over $\boldsymbol{\beta}$ uses the shotgun stochastic search (SSS) algorithm from FINEMAP, but with the likelihood modified to marginalise over $\boldsymbol{\alpha}$. The SSS is divided into epochs, and variance components (σ^2, τ^2) are updated between epochs using a method-of-moments estimator.

11.2.3 SuSiE-inf

For SuSiE-inf, the sparse effects are parameterised as a sum of single effects $\beta = \sum_{l=1}^L \beta_l$ (as in standard SuSiE). Posterior inference uses the variational approximation framework of SuSiE, with the ELBO modified to account for the infinitesimal component. Variance components (σ^2, τ^2) are updated within each iteration using method-of-moments.

11.2.4 Computational Efficiency

All operations are expressed in terms of the eigenvalues and eigenvectors of the in-sample LD matrix, which can be pre-computed. With pre-computed eigen-decomposition, the additional computational cost of the infinitesimal component is modest.

11.2.5 Posterior Inclusion Probabilities

PIPs are computed in the same way as the parent methods:

- **SuSiE-inf:** $\text{PIP}_j \approx 1 - \prod_{l \in \mathcal{L}} (1 - \alpha_{lj})$ (product formula over single-effect components).
- **FINEMAP-inf:** $\text{PIP}_j = \sum_{\gamma \in \Gamma^*} \mathbf{1}(\gamma_j = 1) p(\gamma \mid \mathbf{z}, \mathbf{LD})$ (marginalisation over SSS configurations).

The PIPs refer only to the sparse component β —the infinitesimal effects α are marginalised out and do not contribute to PIPs.

11.2.6 Credible Sets

SuSiE-inf inherits SuSiE’s credible set construction (cumulative- α_l sorting with purity filtering). FINEMAP-inf inherits FINEMAP’s post-hoc credible set procedure. The authors report that SuSiE-inf produces $\sim 42\%$ fewer credible sets than SuSiE, and those that remain are smaller on average.

11.2.7 Recommended Aggregation: minPIP-inf

The authors recommend using the minimum PIP between SuSiE-inf and FINEMAP-inf ($\text{minPIP-inf} = \min(\text{PIP}_{\text{SuSiE-inf}}, \text{PIP}_{\text{FINEMAP-inf}})$) for each variant, as this improves agreement between the two methods and further reduces false discoveries.

11.2.8 Summary of the Pipeline

1. Run GWAS (BOLT-LMM or OLS) to identify genome-wide significant loci.
2. For each locus, pre-compute the eigen-decomposition of the in-sample LD matrix.
3. Run SuSiE-inf and/or FINEMAP-inf, which jointly estimate:
 - Sparse PIPs for each variant (the fine-mapping output);
 - Per-locus infinitesimal variance $\hat{\tau}^2$;
 - Residual variance $\hat{\sigma}^2$.
4. Construct credible sets (inherited from parent method).
5. Optionally: compute $\text{minPIP-inf} = \min(\text{PIP}_{\text{SuSiE-inf}}, \text{PIP}_{\text{FINEMAP-inf}})$.

11.3 Evaluation Metrics

The evaluation of SuSiE-inf / FINEMAP-inf introduces a novel real-data metric (Replication Failure Rate) alongside standard simulation-based calibration and recall metrics.

11.3.1 Replication Failure Rate (RFR) — Novel Metric

The primary real-data metric is the Replication Failure Rate (RFR), defined as:

$$\text{RFR} = \Pr(\text{PIP}_{366\text{K}} < 0.1 \mid \text{PIP}_{100\text{K}} > 0.9), \quad (90)$$

i.e., the proportion of high-confidence variants identified in a 100K subsample that fail to replicate (PIP drops below 0.1) in the full 366K cohort. Under a well-calibrated method, RFR should be close to the Expected Proportion of Non-causal variants (EPN), estimated as $1 - \overline{\text{PIP}}$ among variants with $\text{PIP} > 0.9$.

Observed RFRs: SuSiE 15%, FINEMAP 12%, substantially higher than EPN ($\sim 2\%$). This indicates that standard methods are overconfident. SuSiE-inf substantially reduces RFR.

11.3.2 PIP Calibration (Simulation)

In simulations where ground truth is known, calibration is assessed by binning variants by PIP and comparing the true proportion of causal variants in each bin to the expected proportion (mean PIP). A well-calibrated method shows agreement between the coloured markers (true proportion) and short black lines (expected proportion) in calibration plots. Results are shown for all simulation scenarios (non-sparsity, stratification, ideal).

11.3.3 Recall (Simulation)

Recall is defined as the percentage of simulated large-effect (sparse) causal variants among the top N variants when ranked by PIP, for $N \in \{50, 100, 500, 1000, 5000\}$. This is a rank-based metric similar to FINEMAP’s “proportion of causal SNPs in top- T ” curve. Error bars are 95% Wilson confidence intervals.

SuSiE-inf and FINEMAP-inf show very similar, but slightly lower, recall compared to their parent methods.

11.3.4 Functional Enrichment (Real Data)

In the UK Biobank application, high-PIP variants are categorised into three disjoint functional categories: coding, putative regulatory, and non-genic. Functional enrichment is compared between:

- SuSiE-inf-specific variants (high PIP in SuSiE-inf but not SuSiE);
- SuSiE-specific variants (high PIP in SuSiE but not SuSiE-inf);
- Background (all variants in fine-mapped regions).

SuSiE-inf-specific variants are 58% more enriched in functionally important categories than SuSiE-specific variants ($P = 6 \times 10^{-4}$).

11.3.5 Cross-Ancestry Polygenic Risk Score (PRS) Prediction

A unique validation: posterior mean effect sizes from the sparse component β are used as PRS weights, and prediction accuracy (delta R^2) is assessed across 6 held-out ancestries (AFR, AMR, CSA, EAS, EUR test, MID) and 7 traits. SuSiE-inf/FINEMAP-inf achieve ~ 10 -fold improvement in delta R^2 over SuSiE/FINEMAP.

11.3.6 Credible Set Size Comparison (Real Data)

SuSiE-inf produces 42% fewer credible sets than SuSiE. Among remaining credible sets, those from SuSiE-inf are smaller on average.

11.3.7 Estimated Infinitesimal Variance $\hat{\tau}^2$

The per-locus estimates of τ^2 are examined across traits and across loci. Height shows the highest estimates; LDL shows the lowest. Estimates increase with the number of credible sets per locus. The authors caution against interpreting τ^2 as a direct reflection of trait heritability without further investigation.

11.3.8 Differences from Other Methods' Evaluation

1. **Replication Failure Rate (unique).** No other method introduces a real-data consistency metric based on downsampling. RFR provides a practical assessment of overconfidence without knowing ground truth.
2. **Multi-source misspecification simulation.** The simulation design systematically layers multiple sources of misspecification (non-sparsity, stratification, imputation noise, QC, effect size distribution) and isolates which one drives miscalibration. No other paper does this.
3. **Non-sparsity as the key finding.** The central result—that non-sparse genetic architecture is the primary driver of PIP miscalibration—provides a mechanistic explanation for why methods overperform in simulations but underperform in real data.
4. **Scale.** $N = 149\,630$ for simulations, $N = 366\,194$ for real data, genome-wide. Far larger than the locus-level simulations of SuSiE ($n = 574$), BEATRICE ($n = 5\,000$), FiniMOM ($n = 5\,400$), SparsePro ($N = 353\,570$ but 3 loci only), CARMA ($n = 10\,000$), or Funmap ($n = 20\text{--}50\text{K}$).
5. **BOLT-LMM in the pipeline.** The simulations use BOLT-LMM for GWAS (the standard in UK Biobank analyses), not OLS. This is more realistic but also means stratification effects are partially controlled by the association method.
6. **PRS as fine-mapping validation.** Using cross-ancestry PRS accuracy as a validation metric for fine-mapping quality is unique to this paper.
7. **Effect size prior from real data.** The sparse effect size distribution is estimated from real SuSiE posteriors (Height), not from a parametric assumption. This is more realistic than SuSiE ($\mathcal{N}(0, 0.6^2)$), CARMA ($\mathcal{N}(0, 0.5^2)$), or Funmap ($\mathcal{N}(0, 0.0075/L_0)$).
8. **No AUPRC, no CS coverage/power in simulations.** The standard credible set metrics (coverage, power, size) and AUPRC are not reported in the simulation evaluation. The focus is on PIP calibration and recall.
9. **Relationship to BEATRICE.** BEATRICE independently introduced a noncausal polygenic term g_{NC} in its generative model. SuSiE-inf/FINEMAP-inf formalise this idea as part of the inference procedure (not just the simulation), providing both a diagnosis (RFR) and a solution (the τ^2 term).